# KENYATTA UNIVERSITY
# SCHOOL OF ECONOMICS
# DEPARTMENT OF ECONOMIC THEORY

## EET 200: MICROECONOMIC THEORY II

### 1.0 THE THEORY OF THE CONSUMER

In your elementary microeconomics, the basic principles of consumer behaviour were introduced by laying a strong foundation on the theory of demand on which premise the consumer behaviour is build upon. The consumer behaviour is introduced as a utility maximising behaviour but subject to consumer's ability to purchase. A consumer is portrayed as an agent who goes for the best that he/she can afford.

In the intermediate level we describe more precisely what we mean by 'best' and 'I can afford'. In the first section, we will examine how to describe what a consumer can afford; the next section will focus on the concept of how the consumer determines what is best. We will then be able to undertake a detailed study of the implications of this simple mode of consumer behaviour.

## 1.1 The Budget Constraint

Suppose that there is some set of goods from which the consumer can choose. In real life there are many goods to consume but we will consider two.

Let the consumer's consumption bundle be $(X_1 X_2)$ where $X_1$, represents the number of units the consumer chooses of good 1 and $X_2$ is the number of units of good to be chosen by the consumer.

Let $p_1$ and $p_2$ represent the unit prices for the two goods respectively; and $M$ to represent the amount of money the consumer has to spend.

The budget constraint of the consumer can be written as:

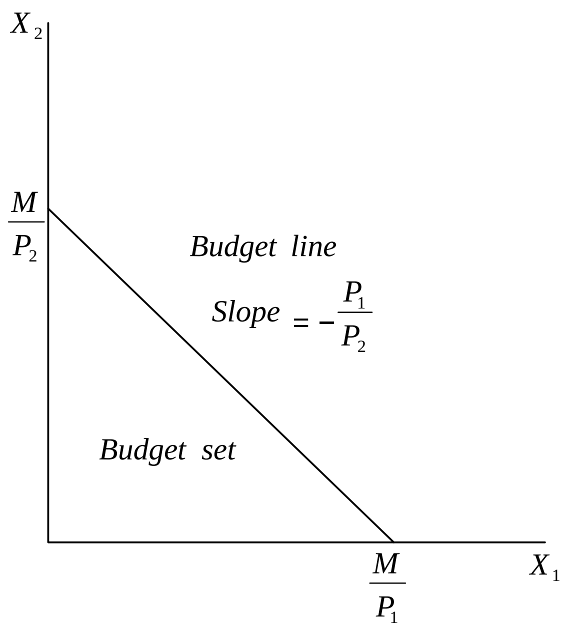
$$P_1X_1 + P_2X_2 \leq M$$

Where $P_1$ $X_1$ is expenditure for good 1 and $P_1X_1$ is expenditure on good 2.

The budget constraint of the consumer requires that the amount of money spent on the two goods is no more than the total amount the consumer has to spend. The consumer's affordable consumption bundles are those that do not costs any more than $M$. The set of affordable consumption bundles at prices $P_1P_2$and income $M$ is called the budget set of the consumer.
The set of bundles that cost exactly M is called the budget line.

$$P_1X_1 + P_2X_2 = M$$

These are the bundles of goods that just exhaust the consumer's income. The budget is set is shown below:




The budget set consists of all bundles that are affordable at the given prices and income.

Re-arranging equations (2)

$$X_2 = \frac{M}{P_2} - \frac{P_1}{P_2} X_2 \text{ ............................................} (3)$$

This is an equation for a straight line with a vertical intercept of $\frac{M}{P_2}$ and a slope of $-\frac{P_1}{P_2}$

The example tells how many units of good 2 the consumer needs to consume in order to just satisfy the budget constraint, if she is consuming $X_1$ units of good 1.

The slope of the budget line measures the rate at which the market is willing to substitute good 1 for good 2.

$$\frac{\Delta X_2}{\Delta X_1} = -\frac{P_1}{P_2}$$

The slope of the budget line is also said to be an opportunity cost of consuming good 1. In order to consume more of good 1, one has to give up some consumption of good 2. Giving up the opportunity to consume good 2 is the true economic cost of more of good 1 consumption, and that cost is measured by the slope of the budget line.

## 1.2 Consumer Preferences

This section is devoted to clarifying the economic concept of "best things" now that the meaning of "can afford" is clear.

The objects of consumer choice are consumption bundles. This is a complete list of the goods and services that are involved in the choice problem faced by a consumer.

Suppose there are two consumption bundles $(X_1 X_2)$ and $(Y_1 Y_2)$. The consumer can rank them as to their desirability. That is, the consumer can determine that one of the bundles is strictly better than the other, or decide that she is indifferent between the two bundles.

### 1.2.1 Assumptions about Preferences

1. **Completeness:** It is assumed that any two bundles can be compared. That is, given any *X* bundle and any *Y* bundle, then $(X_1 X_2)$ $(\geq Y_1 Y_2)$ or $(Y_1 Y_2) \geq (X_1 X_2)$ or both, in which case the consumer is indifferent between the two bundles. This is to say that the consumer can make a choice.

2. **Reflexive:** We assume that any bundle is at least as good as itself

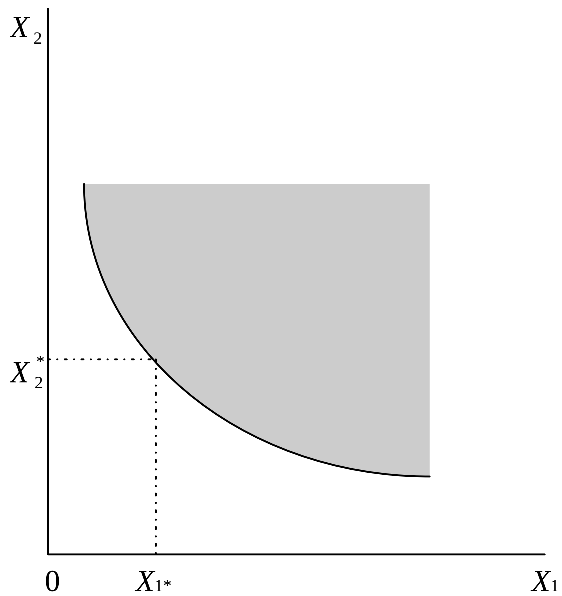
   $(X_1 X_2)$ $(\geq X_1 X_2)$

3. **Transitive:** If $(X_1 X_2) \geq (Y_1 Y_2)$ and $(Y_1 Y_2) \geq (Z_1 Z_2)$ then we assume that consumer thinks that *X* is at least as good as *Y* and that *Y* is at least as good as *Z*, then the consumer thinks that *X* is at least as *Z*.

### 1.2.2 Indifference Curves

The whole theory of consumer choice can be formulated in terms of preferences that satisfy the three axioms above. However, it is convenient to describe preferences graphically using indifference curves.

Consider a consumer's consumption of goods 1 and 2:



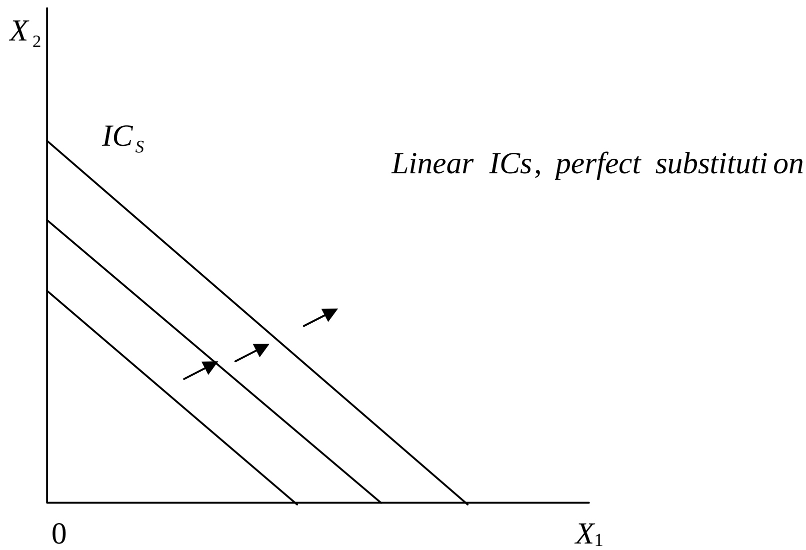

If $(X_1 X_2)$is a certain consumption bundle, the consumption bundle in the shaded region are weakly preferred to $(X_1 X_2)$. It is called the weakly preferred set. The bundles on the boundary of this set for which the consumer is just indifferent to $(X_1 X_2)$ from the indifference curve. It consists of all bundles of goods that leave the consumer indifferent to the given bundle.

### 1.2.3 Shapes of Indifferent Curves

If no further assumptions about preferences are made, ICs can take very peculiar shapes.

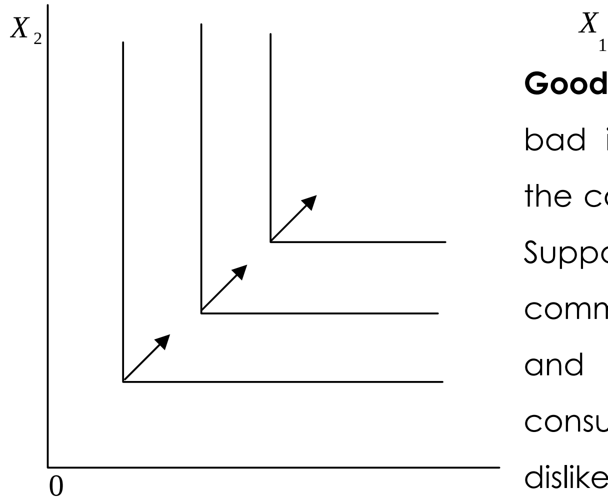
1. **Perfect Substitutes**

   Two goods are perfect substitutes if the consumer is willing to substitute one god for the other at a constant rate. The simplest care of perfect substitutes occurs when the consumer is willing to substitute the goods on a one to one basis. ICs for such a consumer are all parallel straight lines.



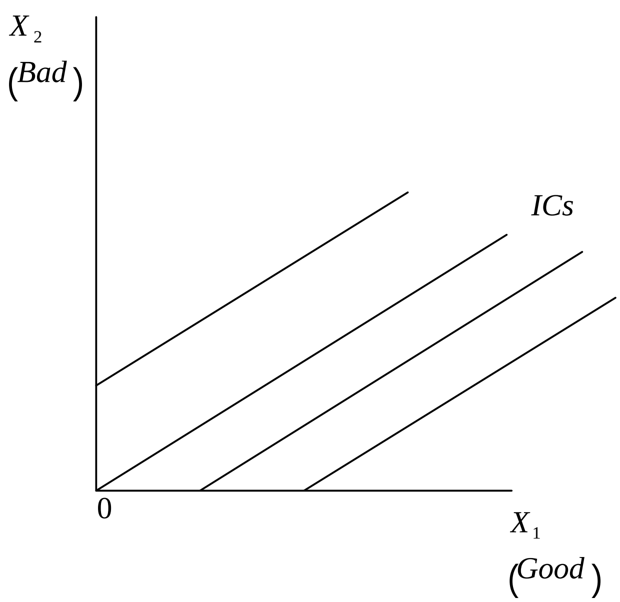

2. **Perfect Complements**

   Perfect complements are goods that are always consumed together in fixed proportions, e.g. shoes (left and right). The ICs are L shaped, with the vertex of the L occurring where the number of one good equals the number of the other good.



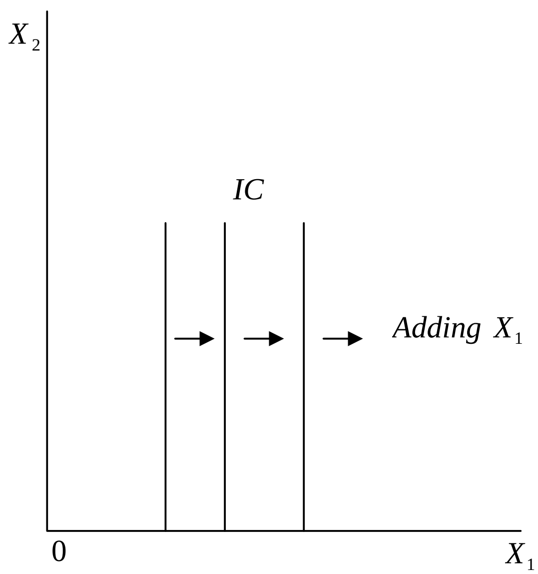

**3. Bad Goods**

A bad is a commodity that the consumer doesn't like. Suppose that the two commodities are meat and pepper, the consumer loves meat but dislikes pepper. But suppose there is some trade off possible between meat and pepper i.e. there would be some amount of meat in samosa that could compensate the consumer for having to consume a given amount of pepper, if more pepper is given in the samosa, more meat has to be given to compensate for having to put up with the pepper. Thus this consumer will have indifference curves that slope up and to the right.



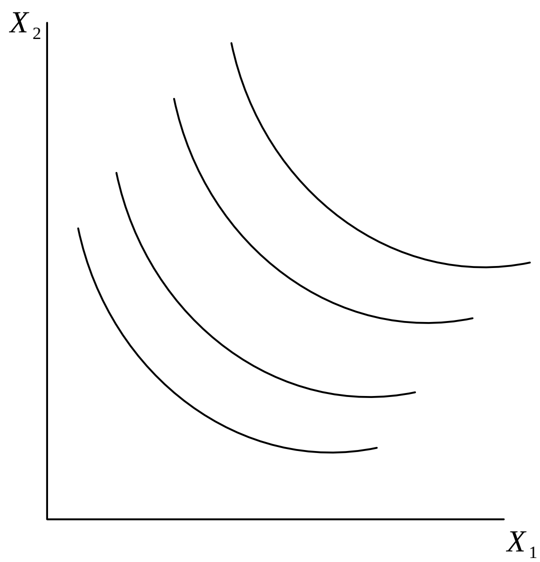
**4. Neutral Goods**

A good is a neutral good if the consumer doesn't care about it one way or the other. Suppose in the above case the consumer is just neutral about pepper $(X_2)$. The IC would be vertical lines as depicted below. The consumer only cares about the amount of $X_1$ and doesn't care at all about how much of $X_2$ he/she has. The more of $X_1$ the better but adding more $X_2$ doesn't affect him.




**5. Imperfect Substitutes**

If the rate at which one good is substituted for another is not constant, but diminishing, then the two goods are imperfect substitutes. As more and more of one good is given up successively larger units of the other good are consumed to compensate the consumer for the loss. Such goods will have indifference curves that are rounded, i.e. the ICs are strictly convex.




### 1.2.4 The Magical Rate of Substitution

The slope of the IC is known as the *MRS*. It measures the rate at which the consumer is just willing to substitute one good for another.

Suppose that we take a little of good 1, $\Delta X_1$ away from the consumer.

Then we give him $\Delta X_2$, an amount i.e. just sufficient to put him back on his/her IC, so that he is just as well off after this substitution of $X_2$ for $X_1$ as he was before.

The ratio $\frac{\Delta X_2}{\Delta X_1}$ is thought as being the rate at which the consumer is willing to substitute good 1 for good 2 and is called the *MRS*.

The *MRS* measures an interesting measure of consumer behaviour. Suppose that the consumer has well behaved preferences, i.e., preferences which are monotonic and convex, and currently consuming some bundle $(X_1 X_2)$. The consumer is now offered a trade: to exchange good 1 for 2 or good 2 for 1 in any amount at a "rate of exchange" of *E*.

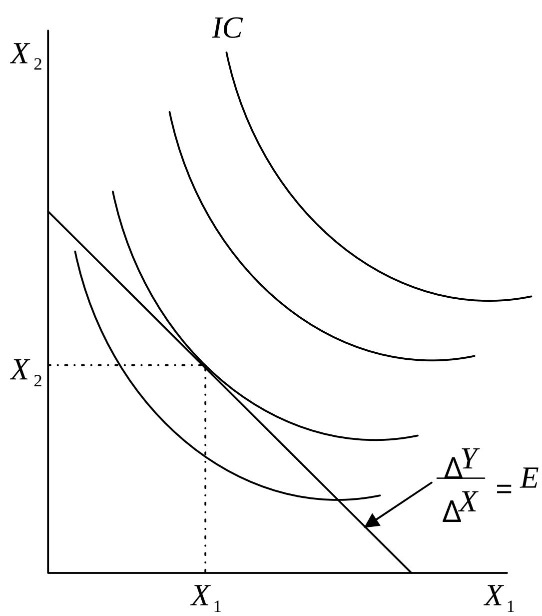
i.e. if the consumer gives up $\Delta X_1$ units of good 1, he can get $E\Delta X_1$ units of good 2 in exchange or conversely, if he gives up $\Delta X_2$, units of good 2, he can get $\frac{\Delta X_2}{E}$ units of good 1.

$$MRS = \frac{\Delta X_2}{\Delta X_1} = E$$

$$\Delta X_2 = E\Delta X_1$$

$$\Delta X_1 = \frac{\Delta X_2}{E}$$

Geometrically, we are offering the consumer an opportunity to move to any point along a line with a slop E that passes through $X_1 X_2$ as depicted.



Moving up and to the left from $X_1$ to $X_2$ involves exchange of good 1 for good 2, and moving down to the right involves exchanging good 2 for good 1. In either movement the exchange rate is *E*. Since exchange always involves giving up one good in exchange for another, the exchange rate E corresponds to slope at *E*.

The point of tangency between the budget line and the indifference curve is referred to as the consumer equilibrium.

### 1.2.5 Behaviour of the Marginal Rate of Substitution

Perfect substitutes indifference curves are characterized by the fact that the *MRS* is constant at -1. The neutral case is characterized by the fact that the *MRS* is everywhere infinite. The preference for perfect complements are characterised by the fact that the *MRS* is either 0 or infinite and nothing in between.

The assumption of monotonicity implies that ICs must have a negative slope, so the MRS always involves reducing the consumption of one good in order to get more of another for monotonic preferences.

The case of convex ICs exhibits yet another kind of behaviour for the *MRS*. For convex ICs, the *MRS* decreases as more of $X_1$ is consumed. Thus the IC exhibits a diminishing *MRS*. Convexity of ICs implies that the more of a good consumed, the more willing is a consumer is to give some of it up in exchange for the other goods. This seems very natural for a consumer and hence convexity if ICs becomes both necessary and sufficient conditions for consumer equilibrium besides just having a point of tangency between the consumer's budget line and the IC.

**1.3 UTILITY**

The theory of consumer behaviour has been formulated with an objective of utility maximization. Utility refers to the ability of/in a good to satisfy the consumer.

There are several approaches to the study of utility with one theory attaching significance to the magnitude utility and is know as the ***cardinal utility*** theory. In this theory the size of the utility difference between the bundles of goods is supposed to have some sort of significance. ]

Another theory formulates the consumer behaviour entirely in terms of consumer preferences and utility is seen only as a way to describe preferences by ranking bundles according to utility derived from each bundle. The proponents of this theory recognized that all that mattered about utility as far as choice behaviour was concerned was whether one bundle had a higher utility than another but how much higher didn't matter (ranking was the only matter). Because of this emphasis on ordering bundles of goods, this kind of utility is referred to as ***ordinal utility***.

**1.3.1 The Utility Function**

A utility function is a way of assigning a number to every possible consumption bundle such that more preferred bundles get assigned larger numbers than less preferred bundles.

A utility function is a way to label indifference curves. Since every bundle of an IC must have the same utility, a utility function is a way to assign number to the different indifference curves in a way that higher ICs get assigned larger numbers. As long as ICs containing more preferred bundles get a larger label than indifference curves containing less preferred bundles, the labelling will represent the different preferences.

However, not all kinds of preferences can be represented by a utility function for example, suppose a consumer had intransitive preferences so that *A>B>C>A*. Then a utility function for these preferences would have to consist of numbers *U(A), U(B)* and *U(C)* such that *U(A)>U(B)>U(C)U(A),* which is impossible.

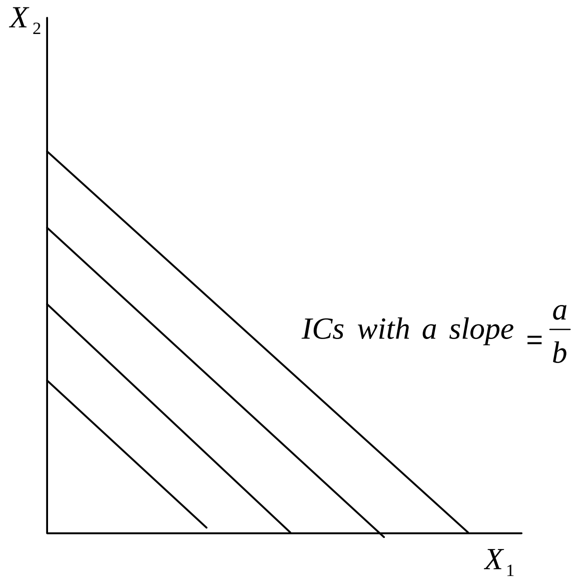
Examples of Utility Functions

1. **Perfect Substitutes goods**

   Preferences for perfect substitutes are represented by linear utility functions of the form.

   $$U(X_1 X_2) = \alpha X_1 + \beta X_2$$

   Where $\alpha, \beta > 0$, numbers that measure the value of goods 1 and 2 to the consumer. By plotting all the points $(X_1 X_2)$ such that $U(X_1 X_2)$ equals a constant, yields a linear indifference curve with a slope $-\frac{\alpha}{\beta}$. For every different value of a constant, there is a different IC, but having a similar slope of $-\frac{\alpha}{\beta}$. Each represents utility functions which are monotonic transformations to each other.



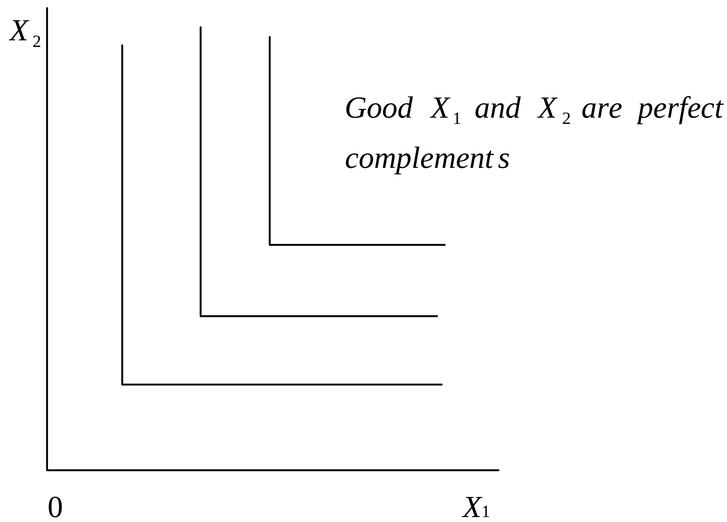

**2. Complementary Goods**

These goods used together. The preferences for complementary goods take utility functions of the form.

$$U(X_1 X_2) = Min(X_1 X_2) = Min(\alpha X_1, \beta X_2)$$

Where $\alpha$ and $\beta$ are positive numbers that indicate the proportion in which the goods are consumed, e.g. $\alpha X_1 < \beta X_2$, then $U(X_1 X_2 = aX_1)$ if $aX_1 > bX_2$ then $U(X_1 X_2) = bX_2$

Any monotonic transformation of this utility function will describe the same preferences which are presented by L-shaped ICs.




**3. Imperfect Substitutes**

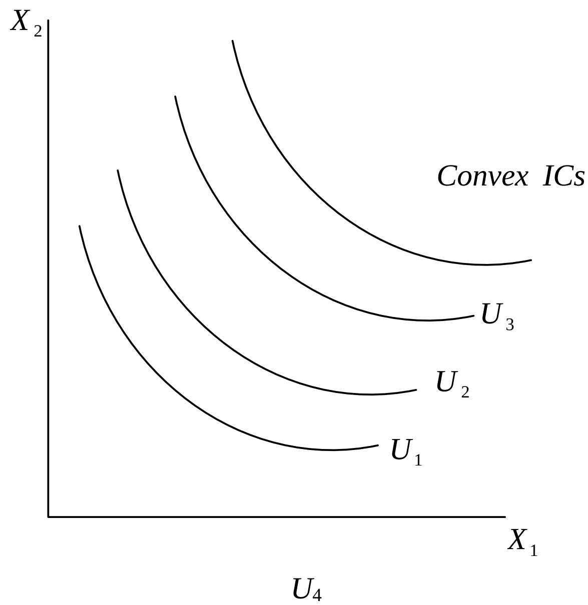
Perfect substitutes have an additive utility function which implies that a consumer will yield some level of utility even by consuming only one of the goods.

However, many cases are such that some level of utility is only possible when a combination of the goods is consumed, i.e. no amount of utility is achievable from consuming zero amount of one good. Such goods are said to be imperfect substitutes and their preferences take multiplicative form of utility functions.

$U(X_1 X_2) = X_1^{\alpha} X_2^{\beta}$

This kind of a function is referred to as a Cobb-douglas utility function. It yields a convex monotonic indifference curve and are said to be well behaved.

It is the most useful preference in presenting algebraic examples of the economic ideas. A monotonic transformation of the Cobb-Douglas utility function above will represent exactly the same preferences as:




**1.3.2 Marginal Utility and Marginal Rate of Substitution**

Consider a consumer consuming bundle $(X_1 X_2)$. The amount by which the total utility changes $U(X_1 X_2)$ when an extra unit of a good is consumed is the marginal utility with respect to the good.

Suppose an extra unit of good 1 is consumed such that total utility changes from $U(X_1 X_2)$ to $U(X_1 + \Delta X_1, X_2)$ the *MU* of good 1 is written as:

$$MU_1 = \frac{\Delta U}{\Delta X_1} = \frac{U(X_1 + \Delta X_1, X_2) - U(X_1 X_2)}{\Delta X_1}$$

It measures the rate of change in utility *(ΔU)* associated with a small change in the amount of good 1$(\Delta X_1)$. Therefore to calculate the change in utility associated with a small change in consumption of good 1, we can multiply the change in consumption by the *MU* of the good.

$$\Delta U = MU_1 \Delta X_1$$

Similarly,

$$MU_2 = \frac{\Delta U}{\Delta X_2} = \frac{U(X_1, X_2 + \Delta X_2) - U(X_1 X_2)}{\Delta X_2}$$

$$\Delta U = MU_2 \Delta X_2$$

Recall that the *MRS* measures the slope of the IC at a given bundle of goods.

Consider now a change in the consumption of each good $(\Delta X_1 \Delta X_2)$ that keeps utility constant, i.e., a change in consumption that moves us along the IC, then

$$MU_1 \Delta X_1 + MU \Delta X_2 = \Delta U = 0$$

Solving for the slope of the IC we have,

$$MRS = \frac{\Delta X^2}{\Delta X_1} = -\frac{MU^1}{MU_2}$$

But $\partial U = 0$ since utility does not change along the indifference curve.

$$MU_{x1}.\Delta X_1 = -MU_{x2}.\Delta X_2$$

$$-\frac{\Delta X_2}{\Delta X_1} = \frac{MU_{x1}}{MU_{x2}}$$

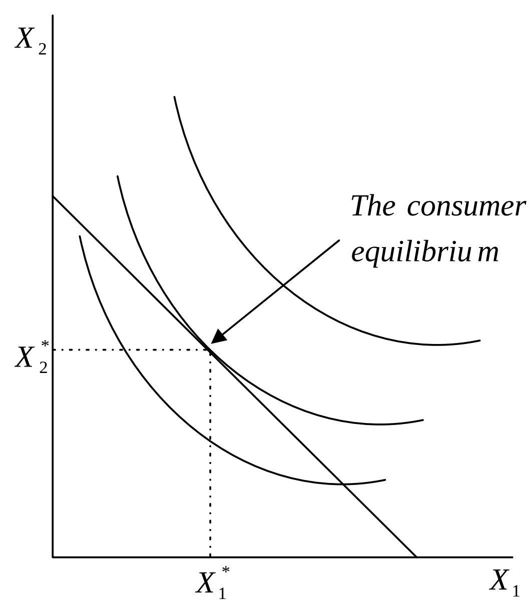
The slope is negative since to get more of one good, lesser units of the other good must be consumed in order to keep the same level of utility. For example if $\frac{\Delta X_2}{\Delta X_1} = -\frac{MU_1}{MU_2} = -2$

Give up 2 units of good 2 to consume extra unit of good 1.

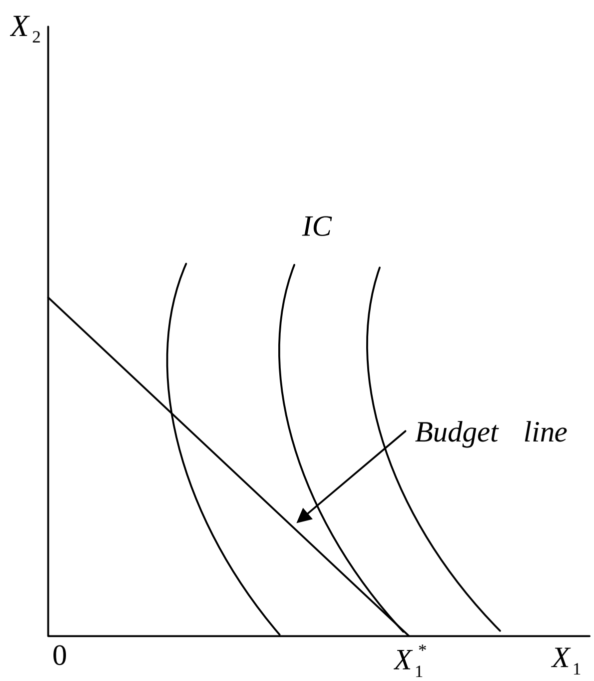
## 1.3.3 Consumer Equilibrium

This section concentrates with putting together the budget line and the theory of preference in order to examine the optimal choice of consumers. We have noted that the economic model of consumer choice is that consumers choose the best bundle they can afford, i.e. the consumers choose the most preferred bundle from their budget set.

Suppose the budget set and the well behaved consumer preferences are drawn on the same diagram as:



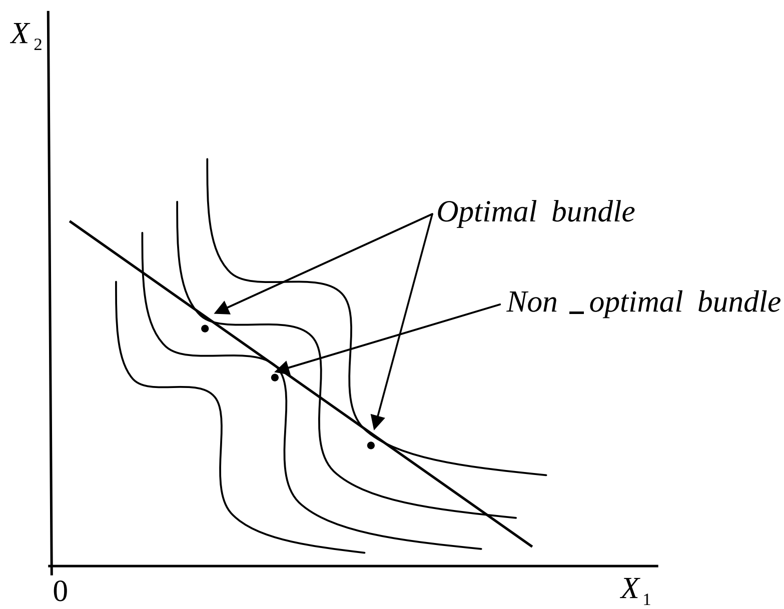
The bundle of goods that is associated with the highest indifference curve and is affordable is $(X_1^* X_2^*)$. Any bundle above this one is unaffordable, such a bundle that appears on a point of tangency between the consumers optimal choice and also referred to as the consumer's equilibrium choice. The point of tangency is called the consumer equilibrium point. Since a solution as is represented above is an interior solution, where the consumers optimal choice involves both goods. A situation where optimal consumption involves consuming 0 units of one good and some units of another is called a boundary solution/optimum. For example:




**Boundary optimum**: The optimum consumption involves consuming 0 units of good 2 and the indifference curve is not tangent to the budget line.

The tangency condition is only a necessary condition to have as optimal choice but not a sufficient condition. However, there is one important case where it is sufficient, the case of convex preferences. In the case of convex preferences, any point that satisfies the tangency condition must be an optimal point.

Generally there may be more than one optimal bundle that satisfies the MRS condition as show below:




In such a case, there are three tangencies but only two optimal points, so the tangency condition is necessary but not sufficient. However, again convexity implies a restriction. If the indifference curves are strictly convex, then there will be only one optimal choice in each budget line.

Recall that the marginal rate of substitution is the rate of exchange at which the consumer is just willing to stay put. On the other hand, the market is offering a rate of exchange to the consumer of $-\frac{P_1}{P_2}$ (the slope of the budget line).

If the consumer is at a consumption bundle where he/she is willing to stay put, it must be one where the *MRS* is equal to this rate of exchange.

$$MRS = -\frac{P_1}{P_2}$$

$$\frac{MU_1}{P_1} = \frac{MU_2}{P_2}$$

Hence

$$MRS = -\frac{MU_1}{MU_2} = -\frac{P_1}{P_2}$$

Or simply

$$\frac{MU_1}{MU_2} = \frac{P_1}{P_2}$$ consumer equilibrium condition

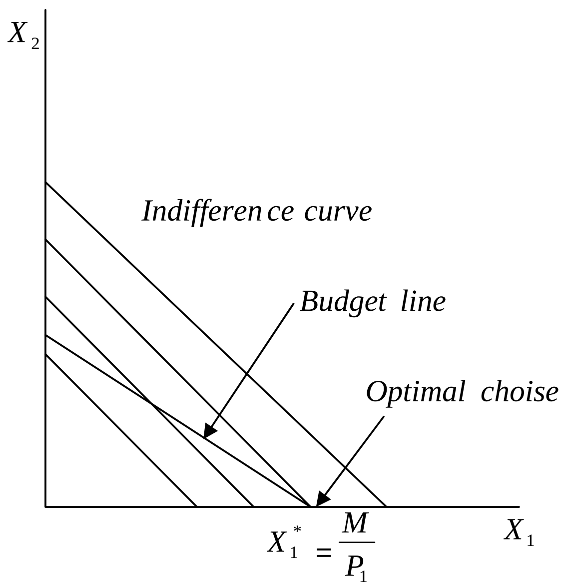
The equilibrium point (optimal choice) has this characteristic that the slope of the IC (*MRS*) is equal to the slope of the budget line. The consumer's choice model can be used to find the optimal choices for other preferences. Some examples include:

**1. Perfect Substitutes**

Recall that perfect substitutes have linear indifference curves, just similar to a budget line. If $P_2 > P_1$, then the slope of the budget line is flatter than the slope of the indifference curves. In such a case, the optimal bundle is where the consumer spends all income on good

1.

If $P_1 > P_2$, then the consumer purchases only good 2. If $P_1 = P_2$, there is a whole range of optimal choices and any amounts of goods 1 and 2 that satisfies the budget constraint is optimal.
The graph below illustrates an optimal choice with perfect substitutes which provide a boundary solution.

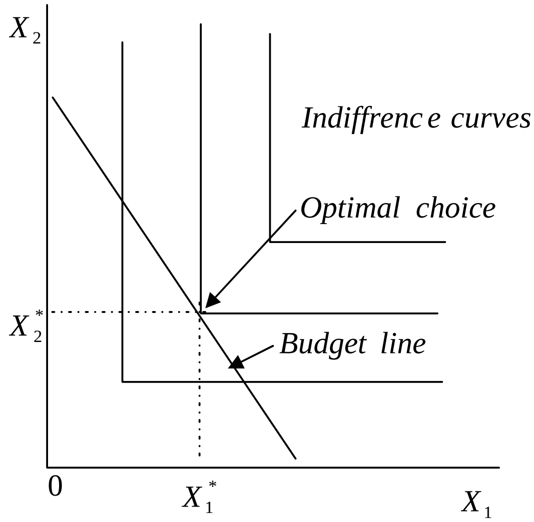



In summury

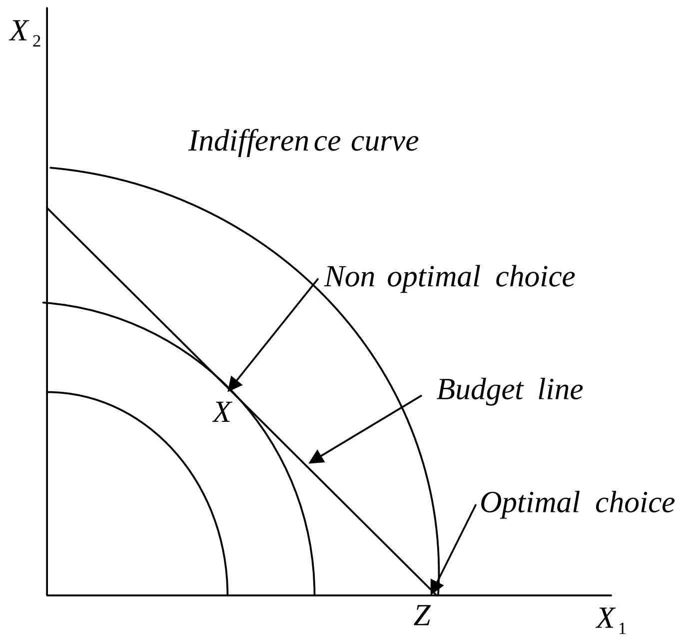
$$X_1 = \begin{cases} \frac{M}{P_1} \; if \; P_1 = P_2 \\ any\ number\ between\ 0\ and\ \frac{M}{P_1} \; if \; P_1 = P_2 \\ 0 \; if \; P_1 > P_2 \end{cases}$$

**2. Perfect Complements**

In this case, the optimal choice must always lie on the diagonal, where the consumer is purchasing equal amounts of both goods, no matter what prices are. The optimal choice can be illustrated as:



**3. Concave preferences**




The optimal choice is always going to be a boundary choice/solution such as bundle z.

The optimal choice is the boundary point z, not the interior tangency point X because Z lies on a higher indifference curve.

## 1.4 The Consumer Demand Functions

The consumer's demand functions give the optimal amounts of each of the goods as a function of the prices and income faced by the consumer. The demand function are written as:

$$X_1 = X_1(P_1 P_2 M)$$

$$X_2 = X_2(P_1 P_2 M)$$

In elementary microeconomics, a clear distinction is made between the normal, inferior and giffen goods.

When the consumer income changes, the optimal choice for the goods changes yielding an income expansion path. The income consumption path is used to derive and Engel curve. On the other hand, a change in price also changes the optimal choice of the consumer yielding a price offer curve which is used to derive the demand curve.

In this level we give it a mathematical approach where we seek to determine the demand functions for the optimal choice bundle of the consumer.

The consumer problem is stated as:

$$MaxU(X_1 X_2)\text{..........................................}(i) \; st$$

$$P_1 X_1 + P_2 X_2 = M\text{......................................}(ii)$$

Introducing a langrangian multiplier and converting (1) and (2) into a composite function we have:

$$L = U(X_1 X_2) - \lambda(P_1 X_1 + P_2 X_2 - M) = 0$$

Obtaining the first order conditions,

$$\frac{\partial L}{\partial X_1} = \frac{\partial U}{\partial X_1} - \lambda P_1 = 0\text{............................................}(iii)$$

$$\partial L \quad \partial U$$

$$\frac{\quad}{\partial X_2} = \frac{\quad}{\partial X_2} - \lambda P_2 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots( )iv$$

$$\frac{\partial L}{\partial \lambda} = P_1 X_1 + P_2 X_2 - M = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(v)$$

Re- writing equations (4) and (5) and dividing them yields:

$$\frac{\frac{\partial U}{\partial X_1}}{\frac{\partial U}{\partial X_2}} = \frac{P_1}{P_2}$$

$$\frac{MU_{X1}}{MU_{X2}} = \frac{P_1}{P_2}$$

This is a similar concept to what was earlier called the consumer equilibrium condition. It corresponds to a point of tangency between an indifference curve and the budget line from the graphical approach discussed in the previous sections.

Solving equations (iii), (iv) and (v) above simultaneously, we obtain the demand functions for the optimal choice bundle. This is best demonstrated using a specific consumer preference and in this case the Cobb-Douglas preference.

Suppose a utility function is specified as

$$U(X_1 X_2) = X_1^{\alpha} X_2^{\beta}$$

The consumer problem is stated as

$$MaxU = X_1^{\alpha} X_2^{\beta}$$

$st$

$$P_1 X_1 + P_2 X_2 = M$$

$$L = X_1^{\alpha} X_2^{\beta} - \lambda(P_1 X_2 + P_2 X_2 - M)$$

$$\frac{\partial L}{\partial X_1} = \alpha X_1^{\alpha-1} X_2^{\beta} - \lambda P_1 = 0 \ldots\ldots\ldots\ldots\ldots\ldots(i)$$

$$\frac{\partial L}{\partial X_2} = \beta X_1^{\alpha} X_2^{\beta-1} - \lambda P_2 = 0 \text{.........................}(ii)$$

$$\frac{\partial L}{\partial \lambda} = P_1X_1 + P_2X_2 - M = 0 \text{.........................}(iii)$$

Re-writing and dividing equation 1 and 2 above we obtain

$$\frac{\alpha X_1^{\alpha-1} X_2^{\beta}}{\beta X_1^{\alpha} X_2^{\beta-1}} = \frac{\lambda P_1}{\lambda P_2}$$

$$\frac{\alpha X_1^{\alpha-1-\alpha} X_2^{\beta-(\beta-1)}}{\beta} = \frac{P_1}{P_2}$$

$$\frac{\alpha X_2}{\beta X_1} = \frac{P_1}{P_2}$$

Such that

$$X_2 = \frac{P_1 \beta}{P_2 \alpha} X_1 \text{...................................}(iv)$$

$$X_1 = \frac{\alpha P_2}{\beta P_1} X_2 \text{...................................}(v)$$

Equation (4) and (5) represent the income expansion paths or the income offer curve.

Substituting equation (4) and (5) into equation (3) one at a time we obtain.

$$P_1X_1 + P_2X_2 = M$$

$$P_1X_1 + P_2\left(\frac{\beta P_1}{\alpha P_2} X_1\right) = M$$

$$P_1 X_1 + \frac{\beta}{\alpha} P_1 X_1 = M$$

$$\left(1+\frac{\beta}{\alpha}\right) P_1 X_1 = M$$

$$\left(\frac{\alpha+\beta}{\alpha}\right) P_1 X_1 = M$$

$$X_1 = \frac{M}{\left(1+\frac{\beta}{\alpha}\right) P_1}$$

$$X_1^* = \frac{M}{\frac{\alpha+\beta}{\alpha} P_1}$$

$$X_1^* = \frac{\alpha M}{(\alpha+\beta) P_1}$$ *demand function for* $X_1$

$$X_2^* = \frac{\beta M}{(\alpha+\beta) P_2}$$ *demand funtion for* $X_2$

$X_1^*$ and $X_2^*$ are referred to as the demand function that represents the optimal choice bundle $X_1^* X_2^*$. They provide the solution to the consumer utility maximising problem.

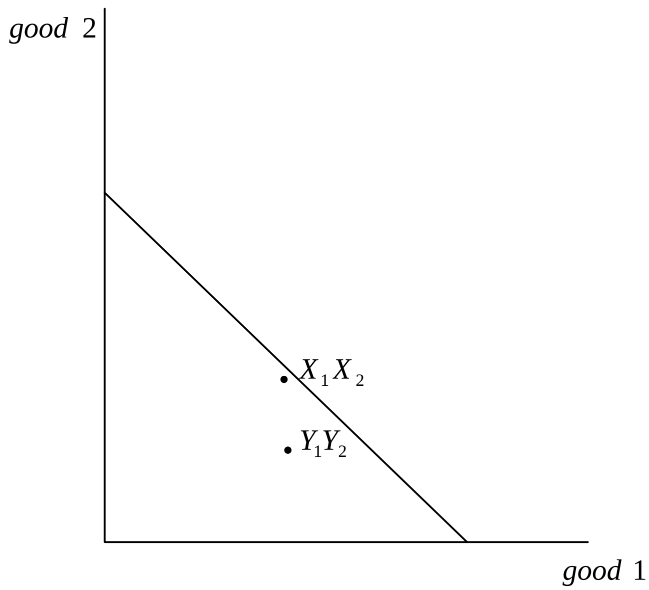
Given the market prices and the income level for a consumer then the functions describe the exact amount of both goods that the consumer would have to consume so as to maximise utility.

## 1.5 Revealed Preference

We have seen how we can use information about the consumer's preferences and the budget constraints to determine his/her demand. However, in real life, preferences are not directly observable. We discover people's preferences from observing their behaviour. The revealed preference theory shows how we can use information about the consumers demand to discover information about his/her preferences.

The theory adopts a maintained hypothesis that the consumers preferences are stable over the time period for which his/her behaviour is observed-whether they may be – are known to be strictly convex. Thus there will be a unique demanded bundle at each budget.

Consider the figure below, where we have depicted a consumer's demanded bundle $(X_1 X_2)$ and another arbitrary bundle $(Y_1 Y_2)$ i.e. beneath the consumer's budget line.




Bundle $(Y_1 Y_2)$is certainly an affordable purchase at the given budget. The consumer could have bought it if he/she wanted to and even had money left over. Since $(X_1 X_2)$ is the optimal bundle it must be better than anything else that the consumer could afford. Hence it must be better than $(Y_1 Y_2)$or any other bundle on or beneath the budget line.

Let $(X_1 X_2)$ be the bundle purchased at prices $(P_1 P_1)$ when the consumer has income M. Since $(Y_1 Y_2)$is affordable at these prices and income, then:

$$P_1Y_1 + P_2Y_2 \leq M$$

Since $X_1X_2$ is actually bought at the given budget, then

$$P_1Y_1 + P_2Y_2 = M$$

Putting these two equations together

$$P_1Y_1 + P_2Y_2 \geq P_1Y_1 + P_2Y_2$$

If the above inequality is satisfied and $(Y_1Y_2)$ is actually a different bundle from $(X_1 X_2)$, then $(X_1 X_2)$ is said to be directly revealed preferred to $(Y_1Y_2)$. This revealed preference is a relation that holds between the bundle that is actually demanded at some budget and the bundles that could have been demanded at that budget. The principle of revealed preference is therefore stated as:-

Let $(X_1 X_2)$ be the chosen bundle when prices are $(P_1P_1)$ and let $(Y_1Y_2)$ be some other bundle such that:

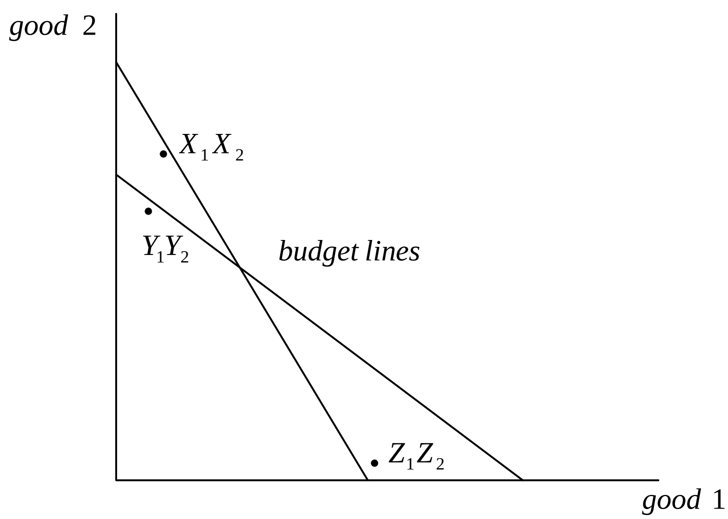
$P_1Y_1 + P_2Y_2 \geq P_1Y_1 + P_2Y_2$. Then if the consumer is choosing the most preferred bundle, he can afford, we must have $(X_1 X_2) > (Y_1Y_2)$.

Suppose further that $(Y_1Y_2)$ is a demanded bundle at prices $(q_1q_2)$and that $(Y_1Y_2)$is itself revealed preferred to some other bundle $Z_1Z_2$. That is:

$$q_1y_1 + q_2y_2 \geq q_1z_1 + q_1z_2 .$$

Then $(X_1X_2)$ $(> Y_1Y_2)$ and$(Y_1Y_1)$ > $(Z_1Z_2)$. From the transitivity assumption, we can conclude that$(X_1X_2)$> $(Z_1Z_2)$.

This is illustrated:




Revealed preference and transitivity assumption tell us that $(X_1X_2)$ must be better than $(Z_1Z_2)$ for the consumer who made the illustrated choices. In this case $(X_1X_2)$ is said to be indirectly revealed proffered to $(Z_1Z_2)$. The chain of direct comparisons can be of any length such that if bundle A is directly revealed to B, and B to C, C to D… all the way to Z, then bundle A is still indirectly revealed to Z.

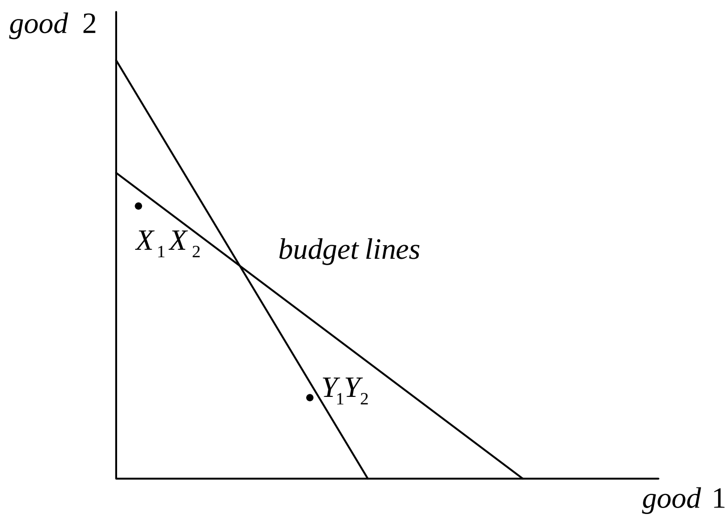
If a bundle is either directly or indirectly revealed preferred to another bundle, we will say that the first bundle is revealed preferred to the second.

From the figure below, since $(X_1 X_2)$ is revealed preferred, either directly, to all of the bundles below (shaded area) either budget lines, $(X_1 X_2)$ is in fact preferred to those bundles by the consumer. That is, the true indifference curve through $(X_1 X_2)$, whatever it is, must lie above the shaded region. It therefore also follows that, the true indifference curve through $(Y_1 Y_2)$ must lie above the flatter budget line.

### 1.5.2 The Weak Axiom of Revealed Preference

We have so far supposed that the consumer has preferences and that he/she is always choosing the best bundle of goods affordable. If the consumer is not behaving this way, the estimates of the indifference curves that we have constructed are meaningless.

Consider:



According to the logic of revealed preference, the diagram allows us to conclude two things.

(i) $(X_1 X_2)$ is preferred to $(Y_1 Y_2)$and

(ii) $(Y_1 Y_2)$ is preferred to $(X_1 X_2)$

That is, the consumer has apparently chosen $(X_1 X_2)$ when he/she could have chosen $(Y_1 Y_2)$ indicating that $(X_1 X_2)$ was preferred to $(Y_1 Y_2)$indicating that $(X_1 X_2)$ was preferred to $(Y_1 Y_2)$ but then he/she indicating the opposite. This situation is absurd and violates the weak Axiom of Revealed Preference.

Clearly this consumer cannot be a maximizing consumer. Either the consumer is not choosing the best bundle he/she can afford or there is some other aspect of the choice problem that has changed that we have not observed. Perhaps the consumer's tastes or some other aspect of his/her economic environment have changed. If consumers are choosing the best bundles that are affordable, but not chosen must be worse than what is chosen. Economists have therefore formulated this simple point in a basic axiom of consumer theory referred to as the weak axiom of revealed preference (WARP) it states that 'if $(X_1 X_2)$ is directly revealed preferred to $(Y_1 Y_2)$ and the two bundles are not the same, then it cannot happen that $(Y_1 Y_2)$ is directly revealed preferred to $(X_1 X_2)$.

In other words, if a bundle $(X_1 X_2)$is purchased at prices $(P_1 P_2)$ and a different bundle $(Y_1 Y_2)$ is purchased at prices $q_1 q_2$ then if.

$P_1X_1 + P_2X_2 \geq P_1Y_1 + P_2Y_2$

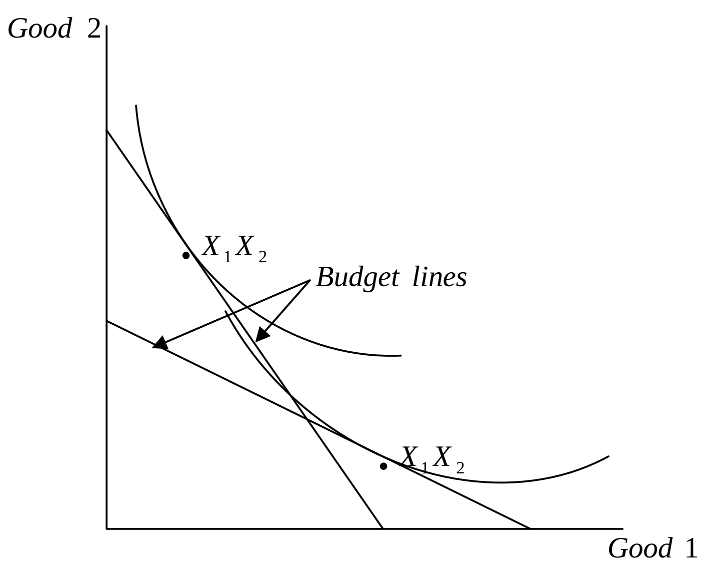
It must not be the case that

$q_1Y_1 + q_2Y_2 \geq q_1X_1 + q_2X_2$

That is, if the Y bundle is affordable when the x-bundle is purchased, then when the Y-bundle is purchased, the X bundle must not be affordable.

The consumer in (immediate diagram) has violated the WARP and therefore this consumers behaviour could not have been maximising behaviour. There is no set of indifference curves that could make both bundles maximizing bundles.

On the other hand, a consumer who satisfies WARP is said to have a maximizing or optimal behaviour and for such, it is possible to find indifference curves for which his/her behaviour is optimal. One possible choice of indifference curves is illustrated below.



## The Strong Axiom of Revealed Preference

The weak axiom of revealed preference requires that if *X* is directly revealed preferred to Y, then we should never observe *Y* being directly revealed to *X*. The strong axiom of revealed preference requires that the same sort of condition hold for indirect revealed preference. The strong Axiom of Revealed Preference. The strong Axiom of Revealed Preference(SARP) states that:

If $(X_1 X_2)$ is revealed preferred to $(Y_1 Y_2)$ either directly or indirectly and $(Y_1 Y_2)$ is different from $(X_1 X_2)$ is revealed preferred to $(Y_1 Y_2)$.

It is therefore clear that if the observed behaviour is optimizing behaviour, then it must satisfy the strong axiom. For if the consumer is optimising and $(X_1 X_2)$ is revealed preferred to, either directly or indirectly, then it must be the case that $(X_1 X_2) > (Y_1 Y_2)$. Therefore, having $(X_1 X_2)$ revealed preferred to $(Y_1 Y_2)$ and $(Y_1 Y_2)$ revealed preferred to $(X_1 X_2)$ would imply that $(X_1 X_2)>(Y_1 Y_2)$ and $(Y_1 Y_2)>(X_1 X_2)$, which is a contradiction. We can conclude that either the consumer must not be optimizing or some other aspect of the consumers environment such as tastes, other prices etc. must have changed.

While WARP is a necessary condition for optimizing behaviour. SARP is both necessary and sufficient in the sense that, if the observed choices satisfy SARP, we can always find well behaved preferences that could have generated the observed choices. It ensures both consistency and transitivity of consumer preferences.

## The Slutsky Equation

Economists are concerned with how consumer's behaviour changes in response to changes in the economic environment. So far, the prices of the goods and the income are the common variables in the consumer economic environment.

The AEC100 an exposure is done on how the optimal choice by a consumer changes due to changes in both prices and income. in this case, we consider a detailed analysis of how a consumer's choice of a good responds to changes in its price.

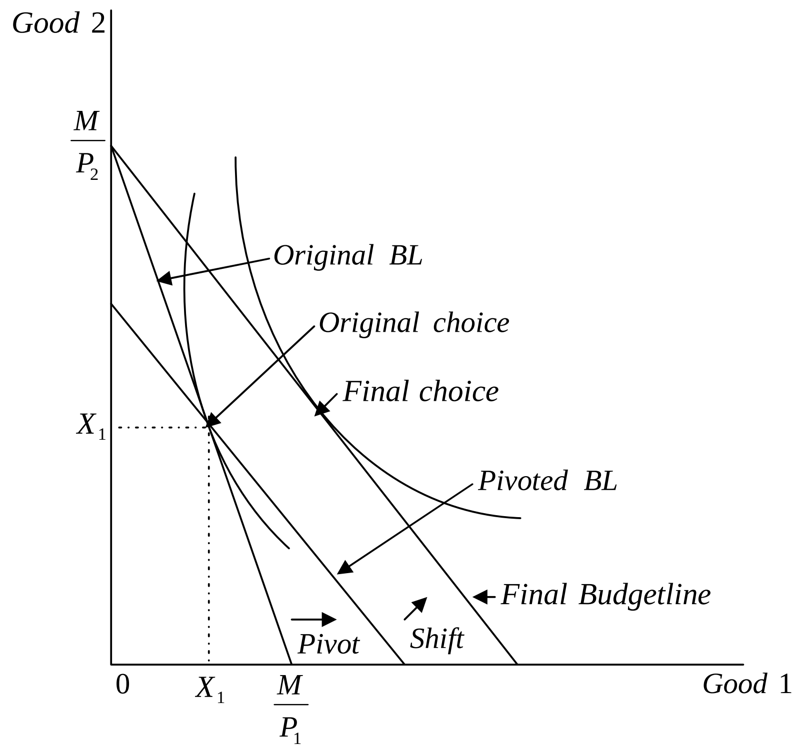
When the price of a good changes, there are two types of effects. The rate at which one can exchange one good for another changes, and the total purchasing power of income changes. These two causes changes in the consumer demand for the goods.

The change in demand due to the change in the rate of exchange between the goods is called the substitution effect. While the change in demand due to changes in the purchasing power is called income effect. To give a more precise definition, let us consider the two effects separately and in details.

**<u>The substitution effect</u>**

To capture the SE, we let the relative prices to change due to a price change and adjust money income so as to hold pursing power constant.

Using the diagram:




Suppose the price of good 1 falls, this means that the budgetline rotates around the vertical intercept $\frac{M}{P_2}$ and becomes flatter. This movement of the budgetline can be broken into two steps:

1. Pivot the budgetline around the original demanded bundle and then,
2. shift the pivoted line out to the new demanded bundle.

The pivot is a movement where the slope of he budgetline changes while its purchasing power remains constant, while a shift is a movement where the slope of the BL remains constant while the purchasing power changes.

These two movements gives a convenient way to decompose the change in demand into two places. This decomposition is only hypothetical and the consumer simply observes a change in price and chooses a new bundle of goods in response. In analysing how the consumers choice changes it is useful to think of the budgetline changing in two stages (i) pivot and (ii) shift.

Considering the pivot, the pivoted budgetline has the same slop and thus the same relative prices as the final budgetline. However, the money income associated with this budgetlines is different since the vertical intercept are different.

Since the original consumption bundle $(X_1 X_2)$ lies on the pivoted BL, that consumption is just affordable. In this sense, the purchasing power of the consumer has remained constant in the sense that the original bundle of goods is just affordable at the new pivoted line. That is after the price of good 1 falls, income must have been adjusted so as to make the original bundle just affordable at a change in relative prices.

Suppose the original prices for goods 1 and 2 are P1 and P2 respectively and M is the original income. Let M1 be the amount of money income that will just make the original consumption bundle affordable this will be the amount of money income associated with the pivoted budgetline.

Since $(X_1 X_2)$ is affordable at both $(P_1 P_2 M)$ and $(P_1{'} P_2 M)$ where $P_1$ is the new fallen price of good 1, then we have

$$M' = P_{1'} X_1 + P_2 X_2$$

$$M = P_1 X_1 + P_2 X_2$$

Subtracting the two equations

$M' - M = X_1 \left( P_1' - P_1 \right)$

That is, the change in money income necessary to make the old bundle affordable at the new price is just the original amount of consumption of good 1 times the change in prices.

Let $\Delta P_1 = P_1' - P_1$. Represent the change in price.

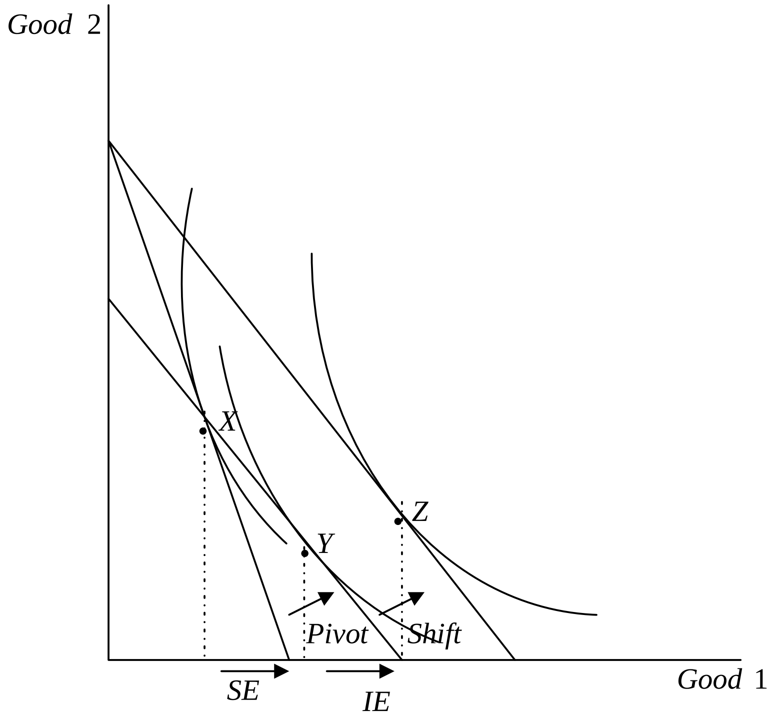
$\Delta M = M' - M$. Represent the change in income necessary to make the old bundle just affordable.

Then:

$\Delta M = X_1 \Delta P_1$

NOTE: The change in income and the changes in price will always move in the same direction: if the price goes up, then we have to raise income to keep the same bundle affordable.

Although $\left( X_1 X_2 \right)$ is still affordable, it is not generally the optimal purchase at the pivoted budgetline. The optimal purchase on the pivoted budgetline is




Bundle Y is the optimal one, when we change the price and adjust income so as to keep the old bundle of goods just affordable. The movement from X and Y is known as SE. It indicates how the consumer substitutes one good for the other. When the price changes while adjusting income to retain the consumer's purchasing power.

More precisely, the SE $\Delta X_1$ is the change in the demand for good 1, when the price of good 1 changes to $P_1'$ from $P_1$ and at the same time, money income changes from M to $M'$

$$\Delta X_1^S = X_1(P_1' M') - X_1(P_1 M)$$

The SE is sometimes called the change in compensated demand. The idea is that the consumer is being compensated for a price change by having his/her income adjusted accordingly, if the price rises "compensation" is done by giving the consumer some more income to make the same amount of a good affordable at a higher price and vice versa.

For example:

**Mathematically**

Suppose that the consumer has a demand function for good x of the form.

$$X = 10 + \frac{M}{10P}$$

Let his original income be Kshs.120 per day and let the price of good X be Ksh.3 per unit. Thus the demand for good X per day is:

$$X(P, M) = X(3,20) = 10 + \frac{120}{10(3)} = 14 \text{ } units\ per\ day$$

Suppose the price of good X falls to Ksh.2 per unit. His new demand at his new price would be

$$X(P', M) = X(2,120) = 10 + \frac{120}{10(2)} = 16 \text{ } units\ per\ day$$

The total change in demand is increased by 2 units of good X per day. In order to calculate the SE, we must first calculate by now much income would have to change in order to make the original demand of 14 units just affordable when the price is Ksh.2 per unit.

Recall that:

$$\begin{aligned} \Delta M &= X_1 \Delta P_1 \\ &= 14(2 - 3) \\ &= -14 \end{aligned}$$

The level of income necessary to keep purchasing power constant is:

$$M' = M + \Delta M$$
$$= 120 - 14$$
$$= 106\ shillings$$

The consumer demand at the new price of Kshs.2 and the new income level of Ksh. 106 is.

$$X(P_1{'} M') = X(2{,}106) = 10 + \frac{106}{10(2)} = 15.3$$

Thus the SE is

$$\Delta X_1{}^S = X(P_1{'} M') - X(P_1 M)$$

**The Income Effect**

We have so far considered the pivot of the budgetline, the second stage of the price adjustment is the shift movement.

A parallel shift of the budgetline is the movement that occurs when income changes while relative prices remain constant. Thus the second stage of the price adjustment is called the income effect. It is a change of the consumers income from M1 back to M1 keeping the prices constant at $(P_1{'}\,P_2)$. In the last diagram this change moves from the point Y to Z. It is natural to call this last movement the income effect since all we are doing is changing income while keeping the prices fixed at the new prices.

Precisely, the income effect $\Delta X_1{}^n$ is the change in the demand for good 1 when the income changes from $M'$ to $M$ holding the price of good 1 fixed at $P_1{'}$

$$\Delta X_1{}^n = X_1(P_1{'} M) - X_1(P_1{'} M')$$

Recall from EC100, that income effect of a price change can operate either way. It will tend to increase or decrease the demand for good 1 depending on whether good 1 is Normal or Inferior.

When the price of a good decreases, the income also decreases in order to keep purchasing power constant. If the good is normal, then this decreases in income will lead to a decrease in demand. If the good is inferior, then the decrease in income will lead to an increase in demand.

For example

$X(P_1' M) = X(2,120) = 16$ $X(P_1'$

$M') = X(2,106) = 15.3$

Thus

$$\Delta X_1^n = X_1(2,120) - X_1(2,106)$$
$$= 16 - 15.3$$
$$= 0.7$$

Since the demand for good X here increases when income increases, then good X is normal.

## <u>The Sign of the Substitution Effect</u>

Consider (last diag) where the amount of good 1 consumed is less that at the bundle X and Y were both affordable at the old prices $(P_1 P_2)$but they were not purchased. Instead, the bundle x was purchased. If the consumer is always choosing the best bundle he/she can afford, then X must be preferred to all of the bundles on the part of the pivoted budgetline that lies inside the original budget set.

That is, the optimal choice on the pivoted budgetline must not be one of the bundles that has underneath the original budgetline. The optimal choice on the pivoted line would have to be either X or some point to the right of X. But this means that the new optimal choice must involve consuming at least as much as good 1 as originally just as we wanted to show. In the (diag.) the optimal choice at the pivoted budgetline is the bundle Y, which certainly involves consuming more of good 1 than at the original consumption point X.

The substitution effect always moves opposite to the price movement. It is said to be always negative, since the change in demand due to the SE is opposite to the change in price. If the price increases, the demand for the good due to the SE decreases.

## The Total Change

The total change in demand is $\Delta X_1$, is the change in demand due to the change in price, holding income constant:

$$\Delta X_1 = X_1\left(P_1' M\right) - X_1\left(P_1 M\right)$$

We have already seen that, this total change can be split into two effects, the substitution effect and $\Delta X_1^S$ and the income effect $\Delta X_1^n$

That is

$$\Delta X_1 = \Delta X_{1S} + \Delta X_{1N}$$

Hence

$$X_1\left(P_1' M\right) - X_1\left(P_1 M\right) = \left[X_1\left(P_1' M\right) - X_1\left(P_1 M\right)\right] + \left[X_1\left(P_1' M\right) - X_1\left(P_1' M'\right)\right]$$

These equations says that the total change in demand equals the substitution effect plus the income effect. This equation is called the Slutsky's identity. It is true for all values of $P_1 P_1^{'}$, M and $M'$. The first and the fourth terms on the RHS cancel out, so the RHS is identically equal to the LHS.

**<u>Use of the Slutsky's Equation</u>**

The Slutsky's identity is not just the algebraic identity. The content comes in the interpretation of the two terms on the RHS. The substitution effect and the income effect. In particular, the signs of the income and substitution effect can be used to determine the sign of the total effect, which in turn reveals the type of a good in question. That is, the total effect reveals whether a good is normal, inferior or giffen.

When the Se must always be negative (opposite to the change in price) the income effect can go either way. Thus the total effect maybe positive or negative.

Normal Good

If a good is normal, the substitution effect and the income effect work in the same direction. An increase in the price will mean that demand will go down due to the SE. If the price goes up, i.e. a decrease in income, which for a normal good means a decrease demand. Both effects reinforce each other. In terms of our notations, the change in demand due to a price increase means that

$$\underset{-}{\Delta X_1} = \underset{-}{\Delta X_{1S}} + \underset{-}{\Delta X_{1n}}$$

On the other hand, if we have an ***inferior good,*** the income effect is positive. A fall in price for example is like an increase in income. an increase in income would reduce the demand for an inferior good.

$$\underset{-}{\Delta X_1} = \underset{-}{\Delta X_{1S}} + \underset{+}{\Delta X_{1n}}$$

However, the second term on the RHS – the income effect – the first term of the right hand side. The total change is therefore negative. This would mean that a fall in price results to an increase in demand.

Giffen good, for a ***giffen good,*** the second term on the RHS – income effect is positive but large enough that the total change could be positive. This would mean that a fall in price would result to a fall in demand.

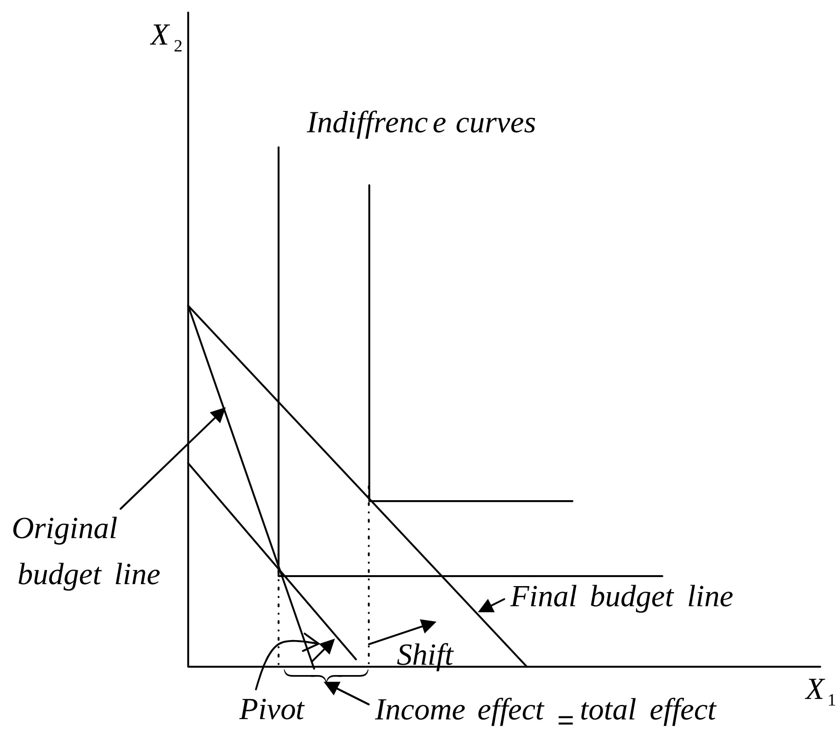
$$\underset{+}{\Delta X_1} = \underset{-}{\Delta X_{1S}} + \underset{+}{\Delta X_{1n}}$$

A fall in price and increased the consumer purchasing power such that, the consumer has reduced his/her consumption of the inferior good. The slutsky's identity shows that his kind of perverse effect can only occur for inferior goods. If the good is normal, then the income and substtituion effect reinforce each other, so that the total change in demand is always in the 'right' directions. Thus a giffen good must be inferior. But an inferior good is not necessarily a giffen good. The income effect not only has to be large enough of the 'wrong' sign it also has to be large enough to outweigh the right sign of the substitution effect. This is why giffen goods are so rarely observed in real life; they would not only have to be inferior goods, but they would have to be too inferior.

**Income and Substitution Effects for Other Preferences.**

A similar analysis could be done for other preferences. A similar analysis could be done for particular kinds of preferences and decompose the demand changes.

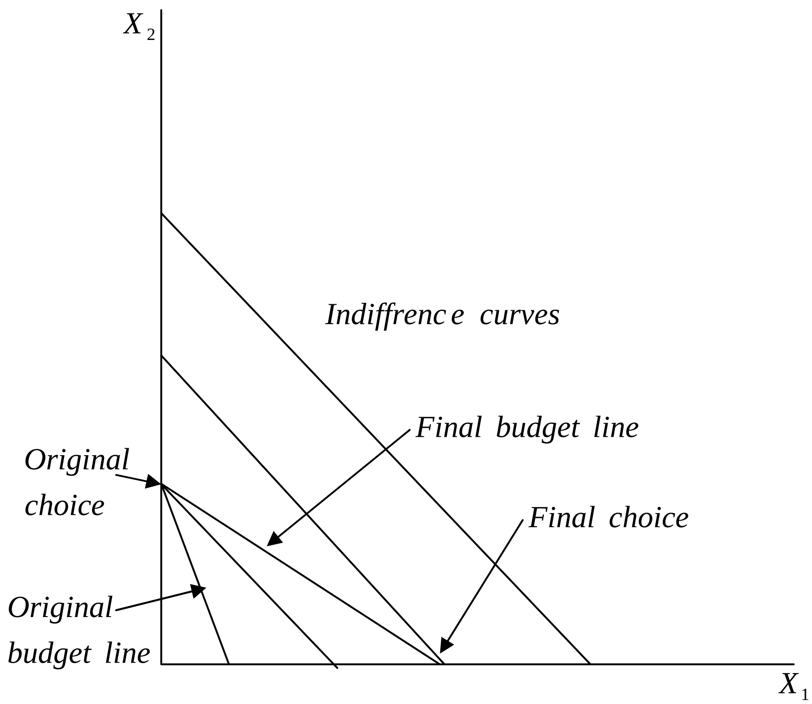
a) perfect compliments:- The slutsky decomposition is shown below:




When the budgetline pivots around the chosen bundles the optimal choice at the new budgetline is the same as at the original one – this means that the substitution effect is zero. The change in demand is entirel due to the income effect.

**b) Perfect substitutes**

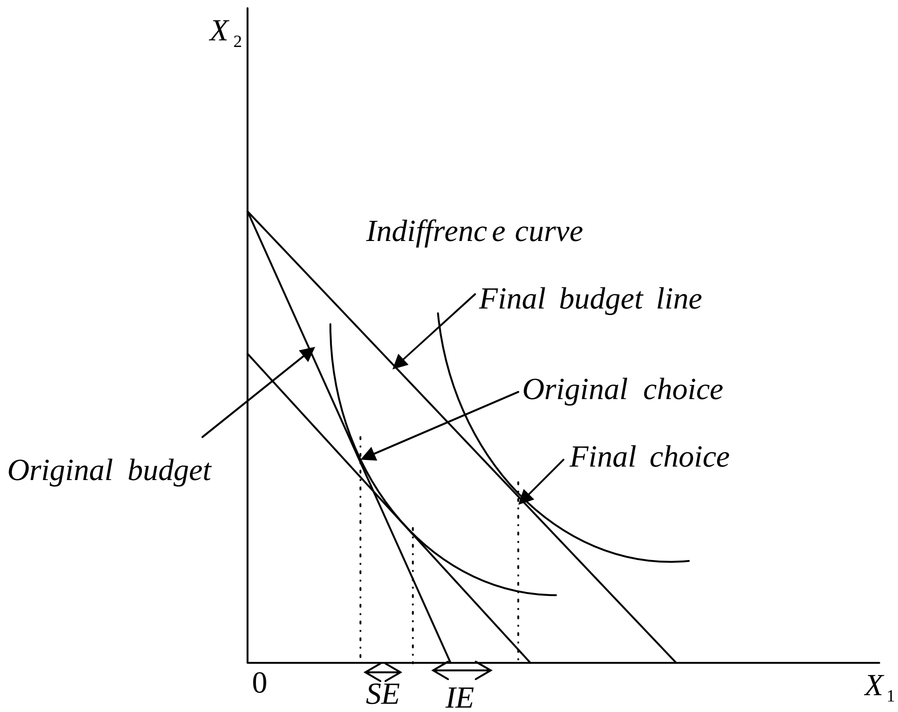
In this case, when the budgetline tilts, the demand bundle moves from the vertical axis to the horizontal axis. There is no shifting and the entire change in demand is due to the substitution effect. This is show as:




**THE HICKS SUBSTITUTION EFFECT**

The substitution effect is the name that economists give to the change in demand when prices change but a consumer's purchasing power is held constant, so that the original bundle remains affordable. This definition is called the slutsky's substitution effect.

Suppose that instead of pivoting the budgetline around the original consumption bundle, the budgetline is rolled around the indifference curve through the original consumption bundle as shown.




The budgetline is pivoted around the indifference curve rather than around the original choice i.e., the consumers purchasing power will no longer be sufficient to purchase his/her original bundle of goods, but will be sufficient to purchase a bund that is just indifferent to his/her original bundle.

Thus the substitution effect keeps the utility level constant rather than keeping the purchasing power constant. This SE is called the Hicks SE.

The Slutsky substitution effect gives the consumer just enough money to get back his/her original level of consumption while the Hicks SE gives the consumer just enough money to get back to his/her original difference curve.

Despite the difference in definition, it turns out that the Hicks SE must be negative in the sense that it is a direction opposite that of the price change just like the Slutsky SE.

## 2.0 THE THEORY OF THE PRODUCER

### Production Technology

We examine the constraints on a firm's behaviour. When a firm makes choices it faces many constraints. Nature imposes the constraints that are only certain feasible ways to produce outputs from the inputs; there are only certain kinds of technological choices that are possible.

### Inputs and Outputs

Inputs to production are called factors of production (classified into broad categories such as land, labour, capital and raw materials). Capital goods are those inputs to production that are themselves produced goods. Basically capital goods are machines of one sort or another.

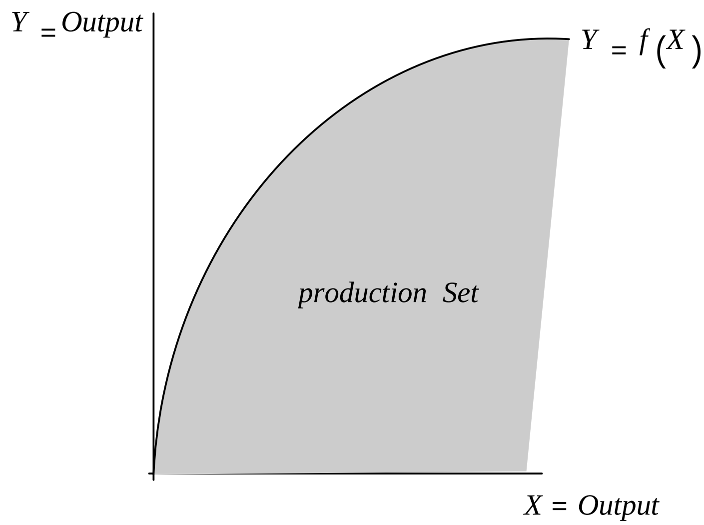
Money used to start up or maintain a business is called financial capital and capital goods or physical capital used for produced factors of production.

### Describing Technological Constraints

Nature imposes technological constraints on firms, only certain combinations of inputs are feasible ways to produce a given amount of output and the firm must limit itself to technologically feasible production plans.

The set of all combinations of inputs and outputs that comprise a technologically feasible way to produce is called a production set.

For example, one input (x) and one output (y) the production set may have the shape




The production set shows the possible technological choices facing a firm. As long as the inputs to the firm are costly it makes sense to limit ourselves to examining the maximum possible output for a production set depicted. The function depicting the boundary set is known as the production function. It measures the maximum possible output that you can get from a given amount of input.

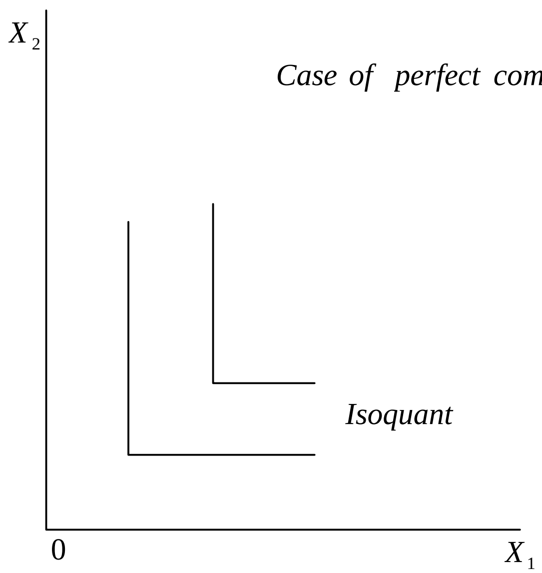
In a two input case $f(x_1 x_2)$ a convenient way to depict productions is by use of isoquant. An isoquant is the set of all possible combinations of inputs 1 and 2 that are just sufficient to produce a given amount of output.

Isoquants are similar to indifference curves, but one difference is that isoquants are labelled with the amount of output they can produce, not with a utility level. Thus the labelling of isoquants is fixed by the technology and does not have the kind of arbitrary nature that the utility labelling has.

**Examples of technology (i) Fixed proportions**

Eg. Producing holes, hence one man one shovel. Extra shovels are not worth anything. Thus the total number of holes that you can produce will be the minimum of the number of men and the number of shovels that you have.

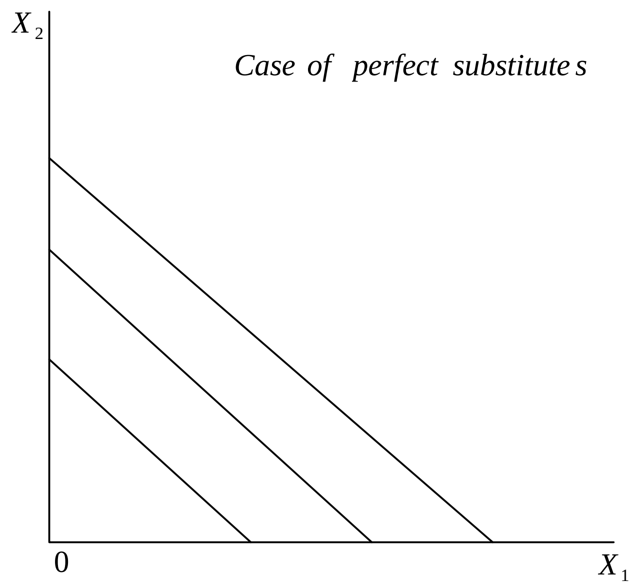
$$f(x_1 x_2) = Min\{x_1, x_2\}$$




**(ii) Perfect substitutes**

Suppose now that we are producing homework and the inputs are red pencils and blue pencils. The amount of homework produced depends only on the total number of pencils, the production function is written as:

$$f(x_1 x_2) = x_1 + x_2$$




**Cobb-Douglas**

A CD production function is given as:

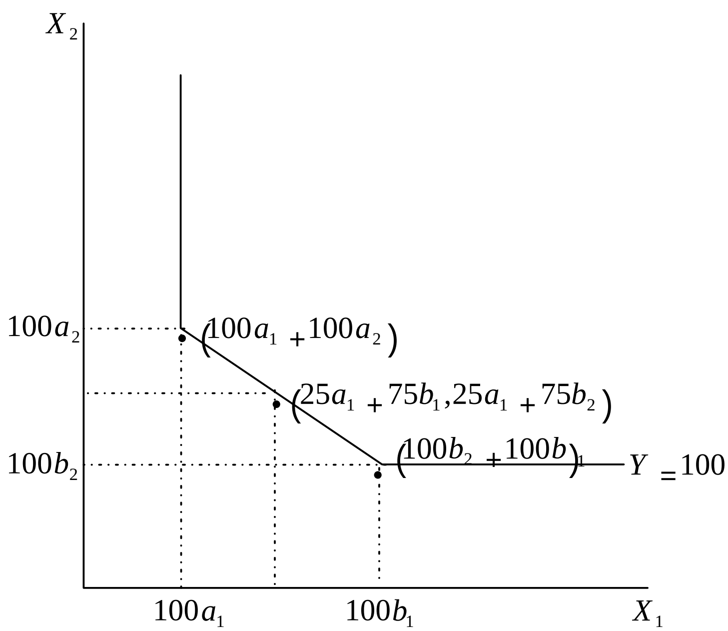
$$f(x_1 x_2) = Ax_1^a x_2^b$$

Measures the scale of production – how much, output we would get if we used one unit of each input. The parameters a and b measure how the amount of output responds to changes in the inputs.

**Properties of technology**

1. Technologies are monotonic – if you increase the amount of at least one of the inputs, it should be possible to produce at least as much output as you were producing originally. This is sometimes referred to as the property of free disposal: if the firm can costlessly dispose of any inputs, having extra inputs around cant hurt it.

2. Technology is convex – this means that if you have two ways to produce y units of output, $(x_1 x_2)$ and $(z_1 z_2)$, then their weighted average will produce at least units of output. Two ways of producing output is called production techniques.




Convexity, if you can operate production activities independently then weighted averages of production plans will also be feasible. Thus the isoquant will have a convex shape.

**The Marginal Product**

Suppose that we are operating at some point, (x1x2) and that we consider using a little bit more of factor 1 while keeping factor 2 fixed at the level x2. How much more output will we get per additional unit of factor 1?

$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2) + f(x_1, x_2)}{\Delta x_1}$$

This is called the marginal product of factor 1. Marginal product is a rate; the extra amount of output per unit of extra input.

**The Technical Rate of Substitution**

Suppose that we are operating at some point $(x_1 \; x_2)$ and that we consider giving up a little bit of factor 1 and using just enough more of factor to produce the same amount of output y. How much extra of factor 2, $\Delta x_2$, do we need if we are going to give up a little bit of factor 1, $\Delta x_1$? This is just the slope of the isoquant referred to as the technical rate of substitution (TRS) denominated by TRS $(x_1 x_2)$.

TRS measures the trade off between two inputs in production. It measures the rate at which the firm will have to substitute one input for another in order to keep output constant.

Consider a change in our use of factors 1 and 2 that keeps output fixed. Then:

$$\Delta y = MP_1(x_1 x_2)\Delta x_1 + MP_2(x_1 x_2)\Delta x_2 = 0$$

Which we can solve to get:

$$TRS(x_1 x_2) = \frac{\Delta x_2}{\Delta x_1} = -\frac{MP_1(x_1 x_2)}{MP_2(x_1 x_2)}$$

**Diminishing Marginal Product**

As long as we have a monotonic technology, we know that the total output will go up as we increase the amount of factor one. But it is natural to expect that it will go up at a decreasing rate. We would typically expect that the marginal product of a factor that will diminish as we get more and more of that factor. This is called the law of diminishing marginal product.

It only holds when other inputs are fixed.

**Diminishing Technical Rate of Substitution**

This assumption says that as we increase the amount of factor 1, and adjust factor 2. so as to stay on the same isoquant, the technical rate of substitution declines. Thus assumption of diminishing TRS means that the slope of an isoquant must decrease in absolute value as we move along the isoquant in the direction of increasing $x_1$. (convex isoquant).

**The Long Run and the Short Run**

In the short run, there will be some factors of production that are fixed at predetermined levels, e.g. land. In the long run all the factors of production can be varied. There is no specific time interval implied here. The long run and the short run periods depends on what kinds of choices we are examining.

Let us suppose that factor 2 is fixed at $x_2$ in the short run. Then the relevant production function for the short run is $f(x_1 x_2)$.

**Returns to Scale**

Instead of increasing the amount of one input while holding the other input fixed, input fixed, let us scale the amount of all inputs up by some constant factor if we get twice as much output, its referred to as constant returns to scale.

$$2f(x_1 x_1) = f(2x_1 2x_2)$$

In general, if we scale all of the inputs by some amount tg, constant returns to scale implies that we should get t times as much output;

$$tf(x_1, x_2) = f(tx_1, tx_2)$$

Returns to scale describes what happens when you increase all inputs, while diminishing marginal product describes what happens when you increase one of the inputs and hold the others fixed. (explained)

If we scale up both inputs by some factor t, we get more than t times as much as output. This is called increasing returns to scale, i.e.

$f(tx_1, tx_2) > tf(x_1 x_1)$ for all $t > 1$

In some instances, we get less than twice as much output from having twice as much of each input. This is called decreasing returns to scale.

$f(tx_1, tx_2) < tf(x_1, x_2)$ for all $t > 1$

**Profit Maximization Short-Run Profit Maximization**

Lets consider the short run profit maximization problem when input 2 is fixed at some level $\bar{x}_2$. Let $f(x_1 x_2)$ be the production function for the firm, let p be the price of output, and let $w_1$ and $w_2$ be the prices of the two inputs. Then the profit maximization problem facing the firm can be written as:

$$\max_{x_1} p.f(x_1 x_2) - w_1 x_1 - w_2 x_2$$

The condition for the optimal choice of factor 1 is not difficult to determine.

If $x_1^*$ is the profit maximizing choice of factor 1 then the output price times the marginal product of factor 1 should equal the price of factor 1.

$$pMP_1(x_1^* \bar{x}_2) = w_1$$

$$MP = w$$

i.e. the value of the marginal product of a factor should equal its price.

The same condition can be described graphically (PTO). The curved line represents the production function holding factor 2 fixed at $\bar{x}_2$. Using y to denote the output of the firm, profits are given by:

$$\pi = py - w_1 x_1 - w_2 \bar{x}_2$$

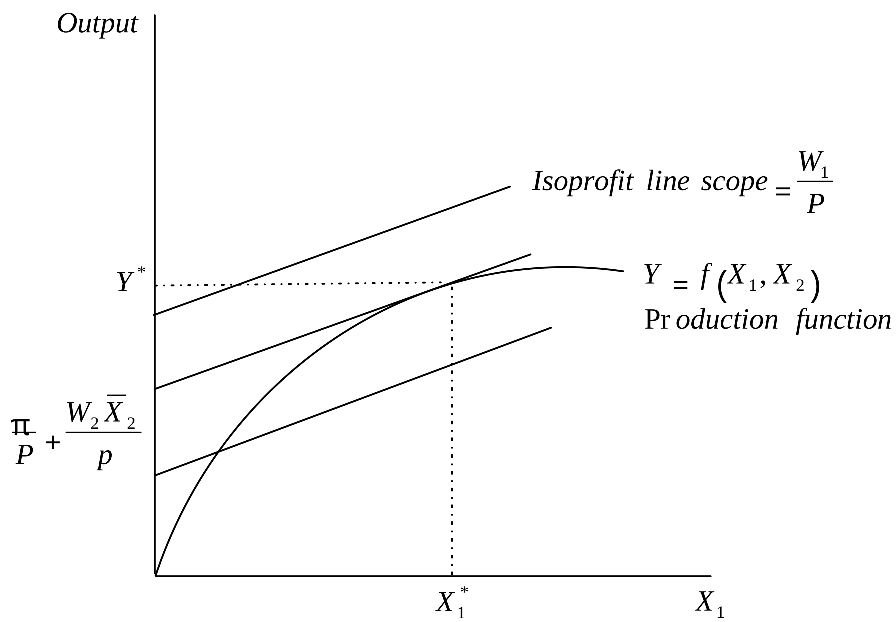
This expression can be solved for y to express output as a function of $x_1$.

$$y = \frac{\pi}{p} + \frac{w_2}{p}\bar{x}_2 + \frac{w_1}{p}x_1 \Big\} \textit{isoprofit line equation}$$

$$\underbrace{\qquad\qquad}_{\text{int}\,ercept}\quad\underbrace{\qquad}_{slope}$$

This equation describes isoprofit lines (combinations of the input goods and the output good that give a constant level of profit,$\pi$). As $\pi$ varies, a family of parallel straight lines each with a slope of w1/p and

each having a vertical intercept of $\frac{\pi}{p} + \frac{w_2\bar{x}_2}{p}$, which measures the profit plus the fixed costs of the firm.




The firm chooses the input and output combination that lies on the highest isoprofit line. In this care the profit –maximizing points is $(x_1^* y^*)$

The profit maximization problem is then to find the point on the production function that has the highest associated isoprofit line. As usual it is characterised by a tangency condition: the slope of the production function should equal the slope of the isoprofit line. Since the slope of the production function is the marginal product, and the slope of the isoprofit line is $\frac{w_1}{p_1}$ this condition can also be written as:

$$MP_1 = \frac{w_1}{p}$$

$$pMP_1 = w_1$$

**Profit Maximization in the Long Run**

In the long run the firm is free to choose the level of all inputs. Thus the LR profit maximization problem can be posed as:

$$\underset{x_1 x_2}{Max}\; pf(x_1 x_2) - w_1 x_1 - w_2 x_2$$

The condition describing the optimal choices is essentially the same as before, to the two factors.

$$PMP_1(x_{1^*} x_{2^*}) = w_1$$

$$pMP_2(x_{1^*} x_{2^*}) = w_2$$

At the optimal choice, the firm's profit cannot increase by changing the level of either input.

**Inverse Factor Demand Curves**

The factor demand curves of a firm measure the relationship between the price of a factor and the profit – maximizing choices: for any prices,

$(p, w_1, w_2)$ we just find those factor demands $(x_1^* x_2^*)$, such that the value of the marginal product of each factor equals its price.

The inverse factor demand curve measures what the factor prices must be for some given quantity of inputs to be demanded.

The profit – maximization problem of the firm is:

$$\max_{x_1 x_2} pf(x_1 x_2) - w_1 x_1 - w_2 x_2$$

Which has FOC

$$\frac{p\partial f(x_1^* x_2^*)}{\partial x_1} - w_1 = 0$$

$$p\frac{\partial f(x_1^* x_2^*)}{\partial x_2} - w_2 = 0$$

Suppose the CD function is given by $f(x_1 x_2) = x_1^a x_2^b$ then the two foc become.

$$pa x_1^{a-1} x_2^b - w_1 x_1 = 0$$

$$pb x_1^a x_2^b - w_2 x_2 = 0$$

Multiply the first equation by $x_1$ and the second by $x_2$ to get.

$$pax_1^a x_2^b - w_1 x_1 = 0$$

$$pbx_1^a x_2^b - w_2 x_2 = 0$$

Using $y = x_1^a x_2^b$ to denote the level of output of this firm we can rewrite these expression as:

$$pay = w_1 x_1 \quad pby = w_2 x_2$$

Solving for $x_1$ and $x_2$

$$x_1^* = \frac{apy}{w_1}$$

$$x_2^* = \frac{bpy}{w_1}$$ *gives demand for the factors as a function of the optimal output choice*

Inserting the optimal to obtain optimal choice of output factor demands into the (i) production function, we have:

$$\left(\frac{pay}{w_1}\right)^a \left(\frac{pby}{w_2}\right)^b = y$$

Factoring out the y gives

$$\left(\frac{pa}{w_1}\right)^a \left(\frac{pb}{w_2}\right)^b y^{a+b} = y$$

$$y = \left(\frac{pa}{w_1}\right)^{\frac{a}{1-a-b}} \left(\frac{pb}{w_2}\right)^{\frac{b}{1-a-b}} \quad \text{Supply function of the CD firm}$$

This gives a complete solution to the profit maximization problem.

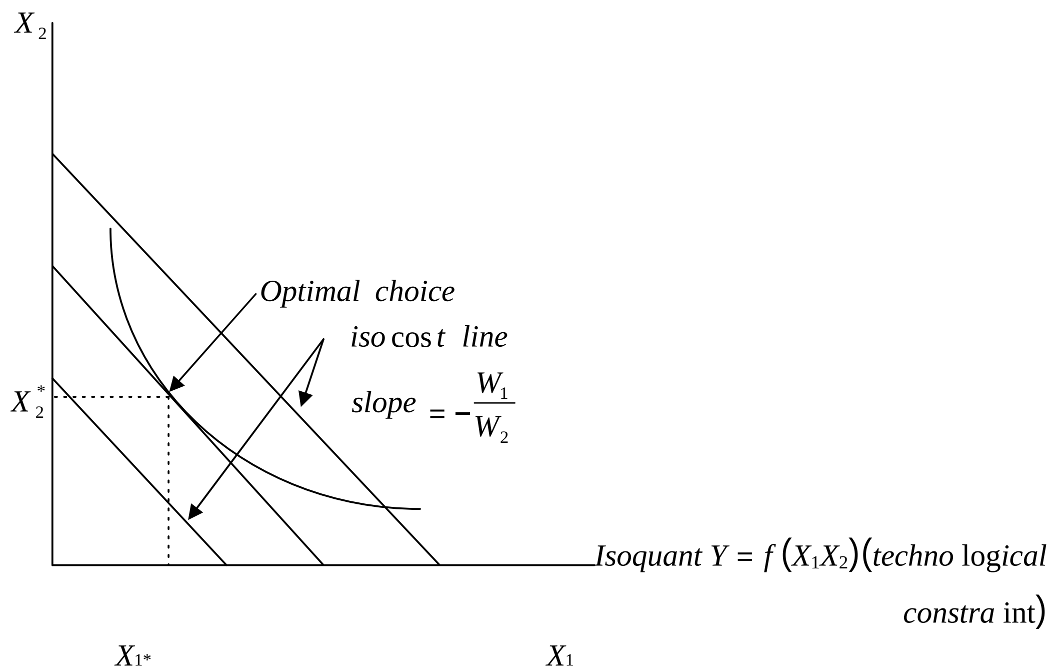
**COST MINIMIZATION**

Suppose that we have two factors of production that have prices $w_1$ and $w_2$ and that we want to figure out the cheapest way to produce a given level of output., y. If we let $x_1$ and $x_2$ measure the amount used of the two factors and let $f(x_1 x_2)$ be the production function for the firm, the problem can be written as:

$$\min_{x_1 x_2} w_1 x_1 + w_2 x_2$$

$$st \quad y = f(x_1 x_2)$$

Minimum cost will depend on w1, w2 and y i.e. $C(w_1, w_2, w_3)$ – cost function measures the minimal costs of producing y units of output when factor prices are $(w_1 w_2)$.




$C = w_1x_1 + w_2x_2$ Rearranging

$$x_2 = \frac{C}{w_2} - \frac{w_1x_1}{w_2}$$

NB: If the optimal solution involves using some of each factor, and if the isoquant is a nice smooth curve, then the cost minimizing point will be characterised by a tangency condition, the slope of the isoquant must be equal to the slope of the isocost curve. i.e. TRS must equal the factor price ratio.

$$-\frac{MP_1(x_1^*x_2^*)}{MP_2(x_1^*x_2^*)} = TRS(x_{1^*}x_{2^*}) = -\frac{w_1}{w_2}$$

The algebra that lies behind equation above

$$MP_1(x_{1^*}x_{2^*})\Delta x_1 + MP_2(x_{1^*}x_{2^*})\Delta x_2 = 0$$

$\Delta x_1$ and $\Delta x_2$ must be of opposite signs, if we are at the cost minimum, then this change cannot lower costs, so we have

$$w_1 \Delta x_1 + w_2 \Delta x_2 \geq 0$$

Now consider the change $(-\Delta x_1, -\Delta x_2)$. This also produces a constant level of output, and it too cannot lower costs. This implies that

$$-w_1 \Delta x_1 - w_2 \Delta x_2 \geq 0$$

Putting the above two expressions together gives:

$$w_1 \Delta x_1 + w_2 \Delta x_2 = 0$$

The choices of inputs that yield minimal costs for the firm will in general depend on the input prices and the level of output that the firm wants to produce written as $x_1(w_1, w_2, y)$ and $x_2(w_1, w_2, y)$.

These are called the conditional factor demand functions or derived factor demands. They measure the relationship between the prices and output and the optimal factor choice of the firm, conditional on the firm producing a given level of output y.

Note difference between conditional factor demands and profit maximization factor demands. Conditional factor demand tells us how much of each factor would the firm use if it wanted to produce a given level of output in the cheapest way. For example

$$MinC = w_1 x_1 + w_2 x_2$$

$st$

$$y = x_1^a x_2^b$$

$$L = w_1 x_1 + w_2 x_2 - \lambda\left(x_1^a x_2^b - y\right)$$

$$\frac{\partial L}{\partial x_1} = w_1 - \lambda a x_1^{a-1} x_2^b = 0$$

$$\frac{\partial L}{\partial x_2} = w_2 - \lambda b x_1^a x_2^{b-1} = 0$$

$$\frac{w_1}{w_2} = \frac{a x_1^{a-1} x_2^b}{b x_1^a x_2^{b-1}}$$

$$\frac{w_1}{w_2} = \frac{a}{b} x_1^{a-1-a} x_2^{b-(b-1)}$$

$$\frac{w_1}{w_2} = \frac{a}{b}\frac{x_2}{x_1}$$

$$w_1 x_1 = \left(\frac{a}{b}\right) w_2 x_2$$

$$x_1 = \frac{a}{b}\frac{w_2 x_2}{w_1}$$

$$x_2 = \frac{w_1 x_1}{w_2}\frac{b}{a}$$

$$x_1 = \frac{a}{b}\left(\frac{w_2}{w_1}\right) x_2$$

$$x_1^{a} x_2^{b} = y$$

$$\left(\frac{a}{b}\frac{w_2}{w_1}\right)^{a} x_2^{a+b} = y$$

$$x_2^{a+b} = \left(\frac{a}{b}\frac{w_2}{w_1}\right)^{-a} y$$

$$x_2 = \left[\left(\frac{b}{a}\frac{w_1}{w_2}\right)^{a} y\right]^{\frac{1}{a+b}} = \left(\frac{b}{a}\frac{w_1}{w_2}\right)^{\frac{a}{a+b}} y^{\frac{1}{a+b}}$$

## **3.0 MARKETS**

1. Competitive Markets

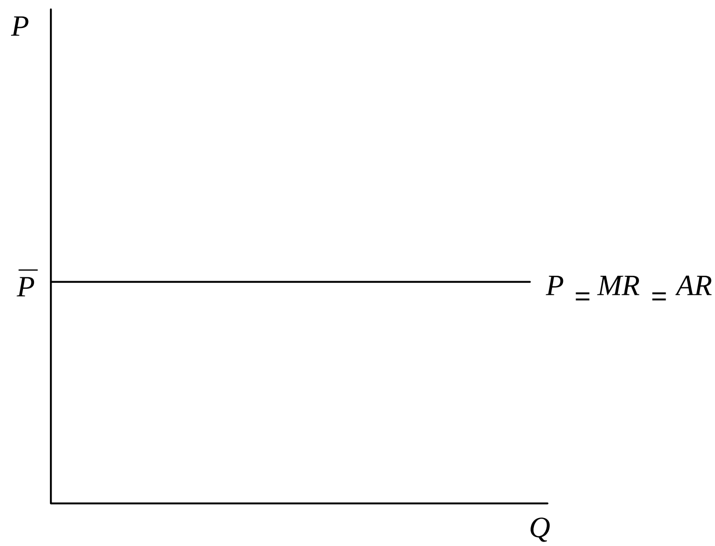
Since the firm is a price taker, the total revenue depends on the amount sold.

$$TR = P \cdot Q$$

$$MR = \frac{\partial TR}{\partial Q} = P$$

$$AR = \frac{TR}{Q} = P$$

$$P = AR = MR$$

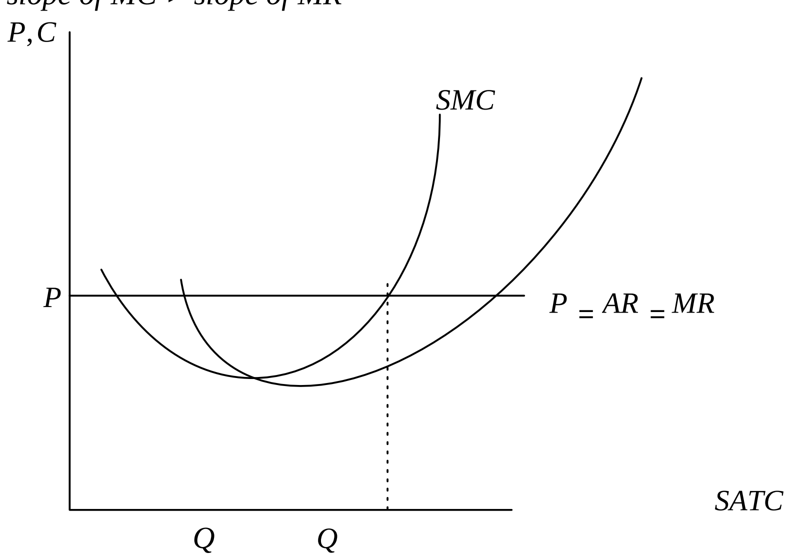



The SMC curve cuts the ATC at its minimum.

At equilibrium

$MR = MC$

$slope\ of\ MC > slope\ of\ MR$



$\pi = TR - TC$

F.O.C. for $\pi$ max

$$\frac{\partial \pi}{\partial Q} = 0$$

$$\frac{\partial \pi}{\partial Q} = \frac{\partial TR}{\partial Q} - \frac{\partial TC}{\partial Q} = 0$$

$$\frac{\partial TR}{\partial Q} - \frac{\partial TC}{\partial Q}$$

$$MR = MC$$

But $MR = P$ for competitive market. At equilibrium $P = MC$

The S.O.C for profit maximisation requires that

$$\frac{\partial^2 \pi}{\partial Q^2} < 0$$

$$\frac{\partial^2 \pi}{\partial Q^2} = \frac{\partial^2 TR}{\partial Q^2} - \frac{\partial^2 TC}{\partial Q^2} < 0$$

$$\frac{\partial^2 TR}{\partial Q^2} < \frac{\partial^2 TC}{\partial Q^2}$$

$$slope\ of\ MR = slope\ of\ MC$$

The MC must be steeper than MR curve hence MC rising.

**2. Monopoly**

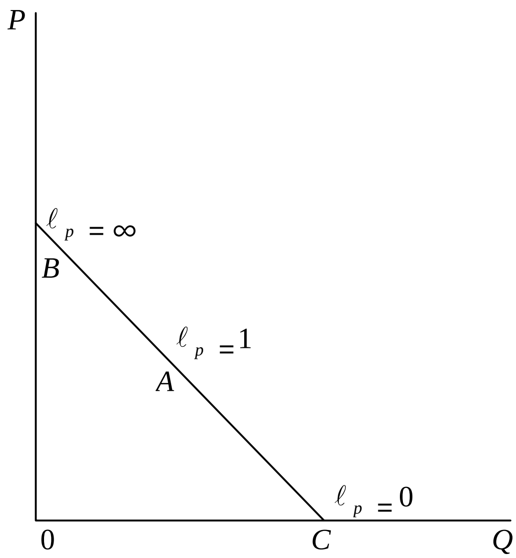
He sets the price hence the amount he sells depend on price, i.e. the demand is a decreasing function of price. i.e.

$$Q = f(p) \quad f' < 0$$

Specifically:

$$Q = b_0 - b_1 p$$

The demand curve is assumed to be linear with changing elastics at every one point.




$$\frac{OQ}{OP} = -b^1$$

$$\ell_p = \frac{OQ}{OP} \cdot \frac{P}{Q}$$

At point B, $\ell_p = -b_1 \cdot \frac{P}{O} = \infty$

At point C, $\ell_p = -b_1 \cdot \frac{0}{O} = 0$

At point A, $\ell_p = 1$

Now

$TR = P \cdot Q$

But $Q = b_0 - b_1 p$

$$P = \frac{b_0}{b_1} - \frac{1}{b_1} Q$$

$$TR = P \cdot Q$$

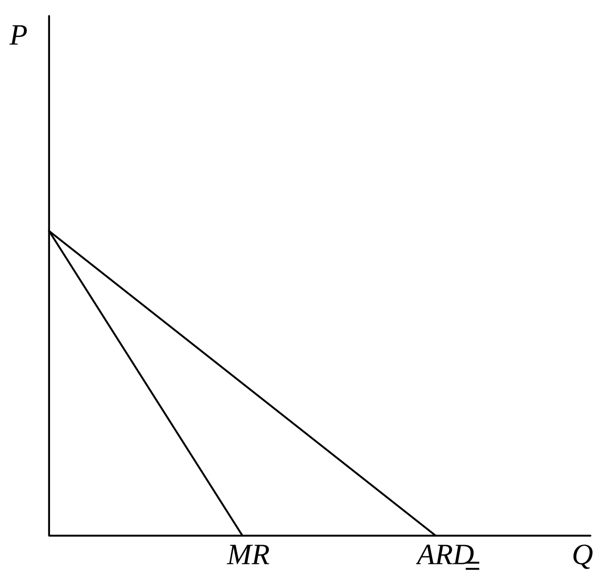
$$= \left( \frac{b_0}{b_1} - \frac{1}{b_1} Q \right) Q$$

$$= \frac{b_0}{b_1} Q - \frac{1}{b_1} Q^2$$

$$AR = \frac{b_0}{b_1} - \frac{1}{b_1} Q$$

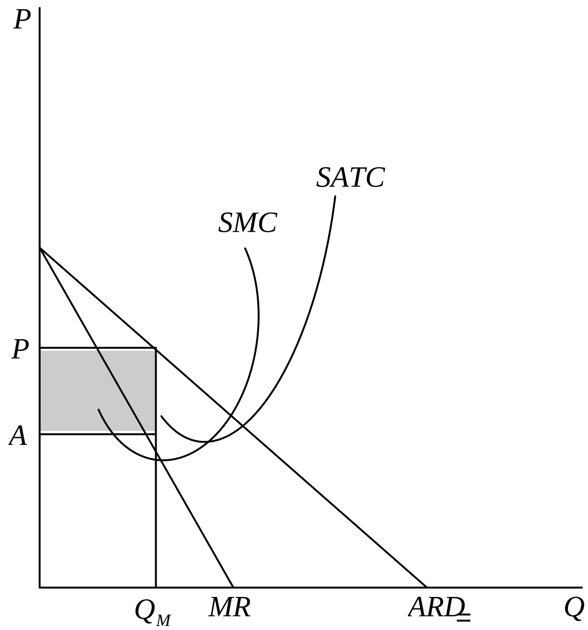
$$MR = \frac{b_0}{b_1} - \frac{2}{b_1} Q$$

The MR is a straight line with the same interrupt as the demand curve but twice as steep as



the AR is the demand curve for a monopolist

Equilibrium requires that MR=MC and slope of MC scope of MC slope of slope of MR at intersection.




$$\pi(Q) = TR(Q) - TC(Q)$$

$$\frac{\partial \pi}{\partial Q} = \frac{\partial TR}{\partial Q} - \frac{\partial TC}{\partial Q} = 0$$

$$MR = MC$$

$$\frac{\partial^2 \pi}{\partial Q^2} = \frac{\partial^2 TR}{\partial Q^2} - \frac{\partial^2 TC}{\partial Q^2} < 0$$

*slope of MR < slope of MC*

e.g.

$$Q = 50 - 0.5P \text{ , } C = 50 + 40Q$$

$$P = 100 - 2Q$$

$$TR = PQ = (100 - 2Q)Q$$

$$= 100Q - 2Q^2$$

$$MR = \frac{\partial TR}{\partial Q} = 100 - 4Q$$

FOC MR=MC

100 – 4Q = 40

60 = 4Q

Q = 15

P = 100 – 2Q

= 100 – 2(15)
=100 – 30

=70 units

SOC $\frac{\partial MR}{\partial Q} = -4, \frac{\partial MC}{\partial Q} = 0$

-4 < 0

**3. The Multiplant Monopolist**

- A monopolist who produces a homegenous product in different plants.
- For simplicity we shall assume two plants.

Assume the monoplast operates two plants A & B with a difference cost structure. He has to make two decisions:

(i) How much output to produce altogether and at what price to sell it so as to maximize profit.

(ii) How to allocate the price of the optimal output between the two plants.

The monopolist is assumed to know his market demand and the corresponding MR curve and the cost structure of the different plants.

The total MC curve is a horizontal summation of the MC curve of the individual plants.

$$MC = MC_1 + MC_2$$

The monopolist maximizes his profit by utilizing each plant upto the level at which the marginal cost are equal to each other and to the common MR. This is *, if the MC of one plant say plant A is lower than the MC of plant B, the monopolist would increase his profit by increasing the production in A and decreasing it in B until

$$MC_1 = MC_2 = MR$$

Given that the market demand

$$P = f(\ )Q = f(Q_1 + Q_2)$$

And the cost structure of the plant

$$C_1 = f(Q_1) \quad C_2$$

$$= f(Q_2)$$

The monopolist aims at the allocation of his production between plant 1 and 2.

To maximise profits

$$\pi = TR - (TC_1 + TC_2)$$

f.o.c

$$\frac{\partial \pi}{\partial Q_1} = 0$$

*and*

$$\frac{\partial \pi}{\partial Q_2} = 0$$

$$\frac{\partial \pi}{\partial Q_1} = \frac{\partial TR}{\partial Q_1} - \frac{\partial TC}{\partial Q_1} = 0$$

$MR_1 = MC_1$

$$\frac{\partial \pi}{\partial Q_2} = \frac{\partial TR}{\partial Q_2} - \frac{\partial TC}{\partial Q_2}$$

$MR_2 = MC_2$

But $MR_1 = MR_2 = MR$ given that each unit of the homogeneous output will be sold at the same price P and will yield the same MR1 irrespective of the plant.

$$\left.\begin{matrix} MR = MC_1 \\ MR = MC_2 \end{matrix}\right\} MR = MC_1 = MC_2$$

S.O.C

$$\frac{\partial^2 TR}{\partial Q_1^2} < \frac{\partial^2 TC}{\partial Q_1^2} \quad \text{and} \quad \frac{\partial^2 TR}{\partial Q_2^2} < \frac{\partial^2 TC}{\partial Q_2^2}$$

$Q = 200 - 2P$

$P = 100 - 0.5Q$

$C_1 = 10Q_1$

$C_2 = 0.25Q_2^2$

$\pi = TR - TC_1 - TC_2$

$TR = PQ$

$= 100Q - 0.5Q^2$

$MR = 100 - Q$

$MR = 100 - (Q_1 + Q_2)$

$MC_1 = 10$

$MC_2 = 0.5Q_2 MR$

$= MC_1$

$100 - Q_1 - Q_2 = 10$

$100 - Q_1 - Q_2 = 0.5Q_2$

$Q_1 + Q_2 = 90 \; Q_1$

$+ Q_2 = 100$

$- 0.5Q_2 = -10$

$Q_2 = \frac{-10}{-0.5}$

$Q_2 = 20$

$Q_1 = 70$

$Q = 20 + 70$

$= 90$

$P = 100 - 0.5(90) = 55$

Obtain profit

**4. Price Discriminating Monopolist**

Exist when the same product is sold at different prices to different buyers.

The conditions necessary which must be fulfilled for the implementation of price discrimination are:

(i) The market must be divided into sub market.

(ii) Different submarkets must have different price elasticities.

(iii) There must be effective separation of the sub markets, so that no reselling can take place from a low price market to a high price market. This condition explains why price discrimination is easier to apply with commodities like electricity or gas or services e.g. doctor which are consumed by the buyer and cannot be resold.

(iv)

**MR Vs price elasticity**

$$TR = PQ$$

$$MR = \frac{\partial TR}{\partial Q} = P\frac{\partial Q}{\partial Q} + Q\frac{\partial P}{\partial Q}$$

$$= P + Q\frac{\partial P}{\partial Q}$$

$$MR$$

but $\ell_p = -\frac{\partial Q}{\partial P}\frac{P}{Q}\,1$

$$\frac{-}{\ell_p} = \frac{\partial P}{\partial Q}\frac{Q}{P}$$

$$\frac{\partial P}{\partial Q} = \frac{1}{\ell_p}\frac{P}{Q}$$

Recall

$$MR = P + Q\frac{\partial P}{\partial Q}$$

$$= P + Q\left[\frac{-1}{\ell_p}\frac{P}{Q}\right]$$

$$= P + \left[\frac{-1}{\ell_p}P\right]$$

$$= P + -P\left[\frac{1}{\ell_p}\right]$$

$$= MR = P\left(1 - \frac{1}{\ell_p}\right)$$

**5. The Impact of Taxes on a Monopolist**

For simplicity let us consider a firm with constant MC. When a quantity tax is imposed the MC will go up by the amount of the tax. What happens to the price?

We consider a linear demand curve.

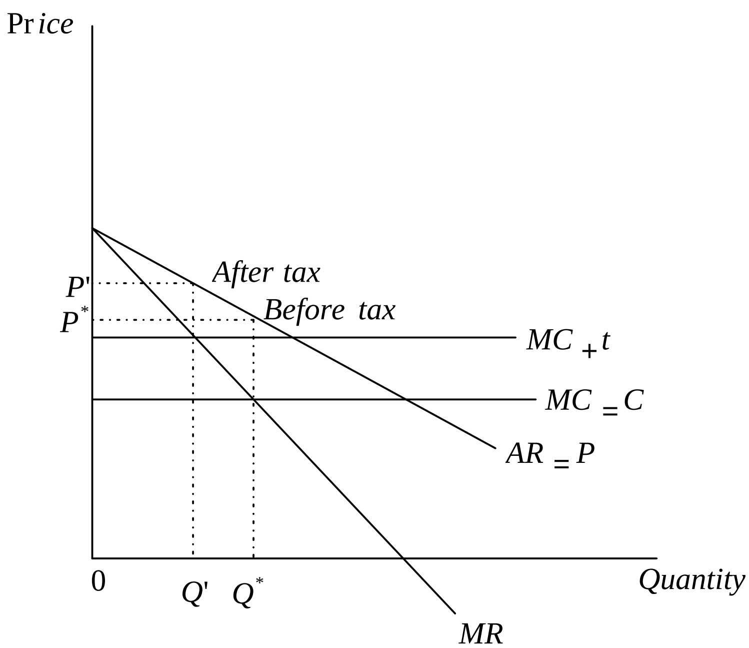
$P = a - by$

Where P is pice

Y is output

$$TR = py = (a - by)$$
$$= ay - by^2 \; MR$$
$$= a - 2by$$

After a quantity tax t, the MC shifts up to the MC + t.




The equilibrium moves to the left after the tax. Since the demand curve is half as steep as the MR curve, the price goes up by half the amount of the tax.

Equilibrium before tax

MR=MC

But after the tax, the MC increases by the text such that:

$MR = MC +$

$t$ But $MC = C$

$a - 2by = c +$

$t \; a - c - t =$

$2by \; 2by = a -$

$c \; -t \; y * =$

$\underline{\qquad\qquad} a - c$

$-t$

$2b$

$$\frac{\Delta y}{\Delta t} = \frac{-1}{2b}$$

Recall that equal price is

$P = a - by^*$

$$\frac{\Delta p}{\Delta y} = -b$$

$$\therefore \frac{\Delta P}{\Delta t} = \frac{\Delta p}{\Delta Y}\frac{\Delta Y}{\Delta t} \Big\} chain\ rule$$

$$= -b \cdot -\frac{1}{2}b = \frac{1}{2}$$

Hence the price changes by less than the tax. Specifically for the linear dd, the price rise by ½ the tax.

Generally: In general, a tax may increase the price by more or less than the amount of the tax.

Recall: $MR = P\left(1 - \frac{1}{|\ell_p|}\right)$

But $MR = MC$

$$P\left(1-\frac{1}{|\ell_p|}\right)=MC$$

$$P=\frac{MC}{1-\frac{1}{|\ell_p|}}$$

$$\frac{\partial P}{\partial t}=\frac{1}{1-\frac{1}{|\ell_p|}}$$

Since the monopolistic will always operate where the demand is elastic then he posses on more than the amount of the tax.

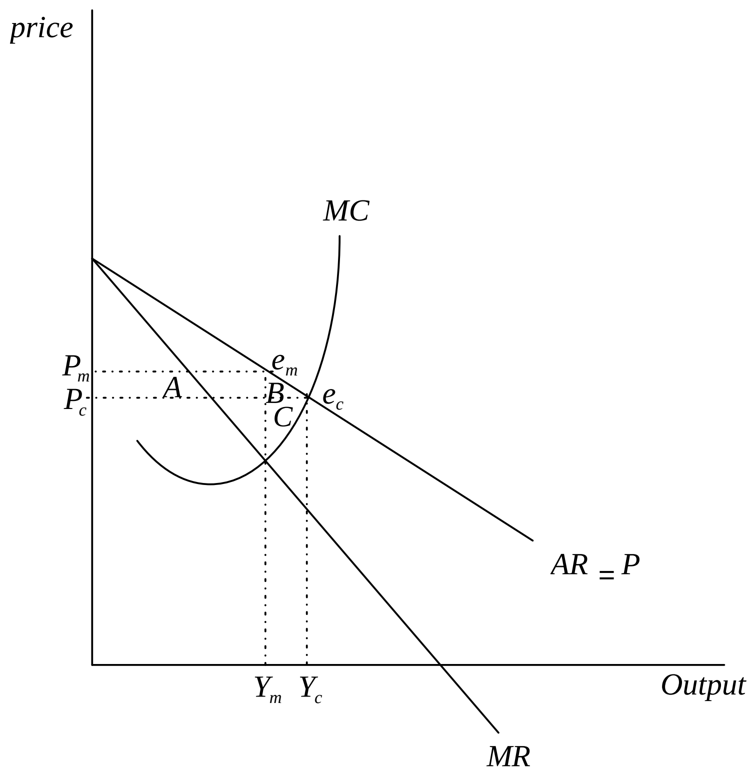
**Inefficiency of Monopoly**

Recall, in a competitive market, P = MC.

In a monopoly, P> MC.

The price will be higher and the output lower, if a firm behaves monopolistically rather than competitively. For this reason, consumers will typically be worse off in an industry organized competitively.

However, the firm will be better off by the same taken into consideration a monopoly situation below:




Suppose that we could somehow costlessly force this firm to behave as a competitor and take the market price as being set exogenously. Then we would have $P_cY_c$ for the competitive price and output.

Alternatively, if the firm recognized its influence on the market price and chose its level of output so as to maximize profits, we would see the monopoly price and output $P_mY_m$ .

Recall that an economic arrangement is pareto efficient if there is no way to make anyone better off without making somebody worse off.

Therefore is the monopoly level of output pareto efficient?

Recall the inverse dd curve *P(Y)* measures how much people are willing to pay for an additional unit of the good. Since *(Y)* is greater than *MC(Y)* for all the output levels between $Y_m$ and $Y_c$, there is a whole range of output where people are willing to pay more for a unit of output than it costs to produce it. Clearly there is a potential for pareto improvement here. Hence a monopoly would be inefficient.

But just how inefficient is he? Can we measure the total loss in efficiency due to monopoly?

The consumers surplus in the monopoly structure is Are $P_m e_m A$ which would go up by area A + B if the monopolist behaved competitively.

On the other hand, the area A would just be a transfer from the monopolist to the consumer.

Under competition, the producer surplus would include area C. However due to the monopoly charging the price Pm, he denies the consumer surplus (A+B). Out of this surplus, only A benefits the producer while B is lost. Area C is also lost since the producer sells $Y_m$ instead of $Y_c$.

The Area B + C is know as the dead weight loss due to the monopoly. It provides a measure of how much worse off people are paying the monopoly price than paying the competitive price.

It measures the value of the lost output by valuing each unit of lost output at the price that people are willing to pay for that unit.

## 4.0 GENERAL EQUILIBRIUM AND WELFARE

The state of efficiency in exchange/consumption i.e. how we can exchange one good for another but maintaining the same level of utility.

We shall assume two people A & B two goods 1 and 2.

$MRS_{12}^{A} = MRS_{12}^{B}$ *Efficiency in exchange*

Efficiency in production where we assume two goods 1 and 2 and two inputs L and K, such that.

$MRTS_{12}^{L} = MRTS_{12}^{K}$ *Efficiency in production*

**The Edgeworth Box Diagram**

This is a convenient graphical tool known as the edgeworth box that can be used to analyze the exchange of two goods between two people. It allows us to depict the endowments and preferences of two individuals in one convenient diagram.

Let us assume two people involved A & B and two goods 1 & 2.

Let person A's consumption bundle be $X_A = \left(X^1_A X_A^2\right)$ where $X^1_A$ represents to A's consumption of good 1 and $X_A^2$ represents As consumption of good 2.

B's consumption bundle of the two goods is $X_B = \left(X_B^1 X_B^2\right)$.

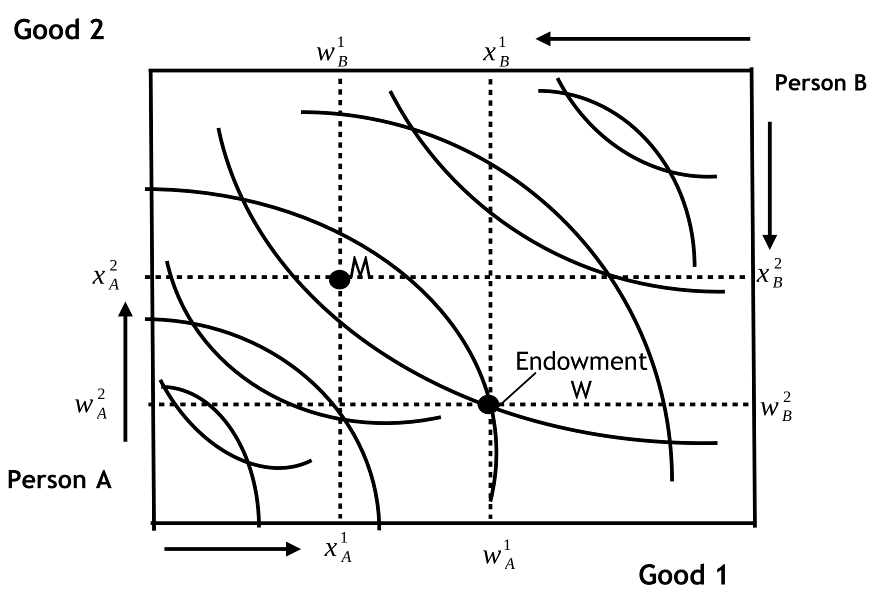
A pair of consumption bundles $X_A$ and $X_B$ is called an allocation. An allocation is a specific allocation if the total amount of each good consumed is equal to the total amount available.

i.e.

$$X_{1A} + X_{B1} = W_{A1} + W_{B1}$$

$$X_{A2} + X_{B2} = W_{A2} + W_{B2}$$

where $W_A^1$ for instance is the total availability of good 1 to person A and $W_B^1$ WB is the availability of good to person B.




The width of the box measures the total amount of good 1 in the economy and the height measures. The total amount of good 2. Persons A's consumption choices are measured from the lower left hand corner while person B's choices are measured from the upper right.

The bundles in this box indicate the amount of goods that each person can hold. If for example there are 10 units of good 1 and 20 units of good 2, then if A holds (7,12), then B must be holding (3,8).

For instance at point M, if person A holds $X^1_A$ if good 1, then B holds $X_B^1$ of good 1. if A holds $X_A^2$ of good 2, then B holds $X_B^2$ of good 2.

**Consider Point W**

Movement from point W to point M for example puts person A to a higher 1C $I_M^A$ from $I_W^A$. Hence making person A better off without necessarily making person B worse off. To move to point M person A have to forego $\left(W_A^1 - X^1_A\right)$ amount of good 1 and have $\left(X_A^2 - W_A^2\right)$ amount more of good 2. on the other hand person B has to forego $\left(W_B^1 - X_B^1\right)$ amount of good 2 in oder to have more of good 1 i.e. $\left(X_B^1 - W_B^1\right)$.

All the same, this movement makes person A better off by putting him on a higher IC. Similarly, movement from point W to point N will make person B better off without making person A worse off.

At a pareto efficient allocation each person is on his highest possible IC given the IC of the other person. A pareto optimal state is therefore a state such that changing the state will at least make one person better off at the expense of the other. One is made better off while the other is made worse off.

A line connecting all such points in consumption contract curve. It is a locus off all efficient allocation in consumption.

A long the c.c.c. the MRS for both persons is the same. When this condition is met, then the first condition for social welfare maximization has been achieved i.e.

$$MRS_{12A} = MRS_{12B}$$

Generally:

$$MRS_{12A} = MRS_{12B} = MRS_{12C} = \dots = MRS_{12n}$$

By definition, a pareto efficient allocation makes each agent as welloff as possible given the utility of the other agent. Let u be the utility level for agent B, so how can agent A be made as well-off as possible?

The maximization problem is:

$$MaxUA\left(X^{1}{}_{A}, X_{A}^{2}\right)$$

$$st$$

$$UB\left(X_{B}^{1}, X_{B}^{2}\right) = \overline{U}$$

$$X_{1A} + X_{B1} = W_{1}$$

$$X_{Z2} + X_{B2} = W_{2}$$

Where.

$W^{1} = W_{A}^{1} + W_{B}^{1}$ is the total amount of good 1 available and $W^{2} = W_{A}^{2} + W_{B}^{2}$ is the total availability of good 2.

We then seek to find $\left(X^1_A X^2_A X^1_B X^2_B\right)$ i.e. $\bar{X}_A$ and $\bar{X}_B$ that makes person B's utility, and given that the total amount of each good used is equal to the amounts available.

We can write the langrangian for this problem as:

$$L = UA\left(X_{1A} X_{A2}\right) - \lambda\left\{UB\left(X_{B1} X_{B2}\right) - \bar{U}\right\} - \mu_1\left(X_{1A} + X_{B1} - W_1\right) - \mu_2\left(X_{A2} + X_{B2} - W_2\right)$$

λ is the language multiplier on the utility constraint and μ's are the language multipliers of the resource constraints.

Getting the FOC's.

$$\frac{\partial L}{\partial X_{1A}} = \frac{\partial UA}{\partial X_{1A}} - \mu_1 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (I)$$

$$\frac{\partial L}{\partial X_{1A}} = \frac{\partial UA}{\partial X_{1A}} - \mu_1 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (II)$$

$$\frac{\partial L}{\partial X_{B1}} = \frac{\partial UB}{\partial X_{B1}} - \mu_1 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (III)$$

$$\frac{\partial L}{\partial X_{B2}} = \frac{\partial UB}{\partial X_{B2}} - \mu_2 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (IV)$$

If we divide the first equation by the second and the third equation by the fourth we have:

$$\frac{\partial U_A / \partial X_A^1}{\partial U_A / \partial X_A^2} = \frac{\mu_1}{\mu_2} \Rightarrow MRS_{12}^A = \frac{\mu_1}{\mu_2}$$

$$\frac{\partial U_B / \partial X_B^1}{\partial U_B / \partial X_B^2} = \frac{\mu_1}{\mu_2} \Rightarrow MRS_{12}^B = \frac{\mu_1}{\mu_2}$$

At a pareto efficient allocation the MRS between the two goods must be the same.